HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

GRADUATION THESIS

Student's engagement detection in online classes

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Program: Data Science and Artificial Intelligence

Supervisor: Trịnh Văn Chiến

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ABSTRACT

Automatically recognising student engagement from video is a key enabler for online and hybrid learning analytics, yet it is hard: engagement labels are *ordinal* (Highly-Disengage, Disengage, Engage, Highly-Engage), severely *imbalanced*, and subjective, and models are rarely tested beyond the corpus they were trained on. This thesis studies engagement classification from per-frame facial-behaviour descriptors (OpenFace) and clip-level video-motion features (I3D), and makes two contributions. A study of monolithic baselines first fixes the evaluation protocol before any model is compared: the six candidate metrics demonstrably disagree on the best model — a disagreement the cross-dataset study later resolves against accuracy — fixing quadratic-weighted kappa (QWK), macro-accuracy, and macro-MAE as the primary metrics; loss choice is shown to be a Pareto trade-off within those primaries, selecting cross-entropy for development; and a clip-level error analysis finds the OpenFace and I3D feature families complementary, motivating their fusion.

First, the thesis proposes a **semantic-group hybrid temporal architecture**. Instead of feeding all 709 OpenFace features into one monolithic sequence model, the descriptor is decomposed into five behaviourally meaningful groups — gaze, eye landmarks, face landmarks, head pose, and action units — and each group is assigned its own temporal encoder (a Temporal Convolutional Network, a Transformer, or an LSTM), optionally fused with an I3D motion stream and supervised by per-stream auxiliary heads. A systematic ablation over all 243 per-group encoder assignments, with and without I3D, shows that the I3D-fused hybrid family is centred above the best monolithic baseline on the CMOSE test set (median quadratic-weighted kappa, QWK, of 0.553 against 0.537; best 0.605), that the strongest configurations mix encoder types, and that the only group with a clear encoder preference is **head pose, which favours a TCN**.

Second, it conducts a **3×3 cross-dataset generalisation study** (training on CMOSE, DAiSEE, or their union; testing on each public test set plus a **self-collected, hand-labelled private set**) and treats the private set as the decisive real-world probe. Engagement classifiers do **not** transfer across corpora — off-diagonal QWK collapses to near zero even when raw accuracy appears acceptable — and the two contributions *compound on the unseen private set*: training on the combined corpus with the semantic-group hybrid yields the best private-set result (QWK 0.379 against 0.285 for the best baseline, a wider margin than in-domain). The thesis argues that order-aware, class-balanced metrics — quadratic-weighted kappa, macro-accuracy, and macro-MAE — not raw accuracy, must be the primary yardstick for imbalanced engagement data.

Keywords: student engagement recognition, affective computing, OpenFace, I3D, temporal convolutional networks, ordinal classification, class imbalance, cross-dataset generalisation.

Student
(Signature and full name)

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LIST OF ABBREVIATIONS

Abbreviation	Definition
AMP	Automatic Mixed Precision
AU	Facial Action Unit
BiLSTM	Bidirectional Long Short-Term Memory network
CDF	Cumulative Distribution Function
CE	Cross-Entropy loss
CMOSE	Class-level Multimodal Online Student Engagement dataset
CNN	Convolutional Neural Network
DAiSEE	Dataset for Affective States in E-Environments
EMD	Earth Mover's Distance (ordinal loss)
GELU	Gaussian Error Linear Unit
I3D	Inflated 3D ConvNet (spatiotemporal video feature extractor)
LSTM	Long Short-Term Memory network
MAE	Mean Absolute Error
MLP	Multi-Layer Perceptron
PCA	Principal Component Analysis
QWK	Quadratic-Weighted Kappa
ReLU	Rectified Linear Unit
RNN	Recurrent Neural Network
SMOTE	Synthetic Minority Over-sampling Technique
SOTA	State Of The Art
SVD	Singular Value Decomposition
TCN	Temporal Convolutional Network

CHAPTER 1. INTRODUCTION

1.1 Problem Statement

Online and hybrid education has become a permanent component of modern teaching. Lectures, tutorials, and assessments are routinely delivered through video conferencing and learning-management platforms, and a large share of learning now happens with a webcam as the only channel between the learner and the instructor. This shift has created both an opportunity and a difficulty. The opportunity is that the webcam stream contains rich behavioural evidence of how a learner is attending to the material; the difficulty is that, unlike in a physical classroom, the instructor cannot read the room. A teacher facing forty live video tiles has no realistic way of noticing that one learner has quietly disengaged ten minutes ago. *Automatic engagement recognition* aims to close this gap: it maps a short video of a single learner to an estimate of how engaged that learner is, so that instructors, tutoring systems, and adaptive platforms can intervene, re-pace a lesson, or measure learning experience at a scale that manual observation can never reach.

This thesis studies the concrete form of that problem that is used by the research community and by the datasets we build on. The input is a short video clip of one learner, on the order of ten seconds, recorded from a frontal webcam. The output is one of four *ordinal* engagement levels: **Highly-Disengage (HD)**, **Disengage (DE)**, **Engage (EG)**, and **Highly-Engage (HE)**. These four levels form a graded scale from complete withdrawal to active, focused participation, and the task is therefore a four-class ordinal classification problem rather than an unordered, nominal one.

Although it resembles ordinary video classification, engagement recognition is harder than a typical image- or action-recognition task for three reasons that recur throughout this thesis and shape every design decision in it.

1. **The labels are ordinal, not nominal.** The four classes lie on a line: HD $<$ DE $<$ EG $<$ HE. Predicting *Highly-Engage* for a *Highly-Disengage* learner is a serious error, whereas predicting *Engage* for a *Highly-Engage* learner is almost right. Plain accuracy ignores this structure entirely, treating both mistakes as equally wrong. Consequently accuracy is the wrong training objective and the wrong evaluation metric; the severity of an error must be weighted by how far apart the true and predicted classes are on the scale.

2. **The labels are severely imbalanced.** Real classes are mostly attentive, so engagement corpora are dominated by the middle-to-upper levels. In the CMOSE

dataset roughly 69% of clips are labelled *Engage* and fewer than 3% are *Highly-Disengage* (Section 3.2, Table 3.1). A model can therefore reach a high accuracy by collapsing onto the majority class and never predicting disengagement at all — precisely the rare, decision-relevant outcome that a useful monitoring system exists to catch.

3. **Engagement is subjective and corpus-specific.** Engagement is not a physical quantity with a ground truth; it is an annotator judgement made under a particular rubric, recording setup, and population. Different datasets differ in all three, so a model trained on one corpus is exposed to a target distribution it has never seen the moment it is deployed. How well models survive this shift is rarely measured, even though deployment always means an unseen distribution.

The combination of these three properties is what makes the problem both practically important and technically interesting. A system that is accurate on paper but blind to disengagement, or that works on the corpus it was trained on but collapses on real footage, is of no use in a classroom. The central concern of this thesis is therefore not headline accuracy but *ordinal agreement on imbalanced data, and its robustness under distribution shift*.

1.2 Background and Problems of Research

Research on vision-based engagement recognition has, broadly, taken two routes. The first encodes each video frame with a facial-behaviour toolkit — most commonly OpenFace [3], which estimates gaze direction, head pose, facial landmarks, and action-unit intensities — and feeds the resulting descriptor sequence to a classical classifier or a recurrent network, for example an OpenFace+BiLSTM pipeline [4] or classical models such as logistic regression, support vector machines, and gradient boosting on aggregated features [5]. The second route learns directly from pixels with end-to-end deep video models, for instance an EfficientNetV2 backbone followed by an LSTM [6], or borrows context- and relationship-modelling ideas from affect recognition [7]. The two large benchmarks used in this thesis sit at the centre of this literature: CMOSE [1] is a recent, large, expert-labelled multimodal corpus that ships precomputed OpenFace and I3D features, and DAiSEE [2] is a widely used public counterpart recorded "in the wild". A systematic review of computer-vision engagement detection [8] catalogues these pipelines and confirms that class imbalance and label subjectivity are recurring, unsolved problems across the field.

Two specific limitations of the existing work motivate this thesis.

First, facial-behaviour descriptors are heterogeneous but are encoded monolithically. A single OpenFace frame in CMOSE is a 709-dimensional vector that bundles together signals of very different physical natures: a handful of gaze direction components, hundreds of eye- and face-landmark coordinates, head-pose translations and rotations, and action-unit intensities. These subsystems move on different time scales — micro-saccades of the eyes are fast and noisy, head turns are slow and smooth, action units fire in brief bursts. The standard practice of concatenating all 709 numbers and pushing them through one sequence encoder forces a single temporal inductive bias onto signals that do not share one. It also makes the model harder to interpret, because the contribution of each behavioural channel is no longer separable.

Second, cross-dataset generalisation is almost never quantified. The engagement literature reports in-domain numbers — train and test on the same corpus — almost exclusively. Yet the only deployment that matters is on data the model has never seen, drawn from a different population, camera, and lighting than the training corpus. Because engagement labels are subjective and the class priors differ sharply between corpora (CMOSE is *Engage*-dominated while DAiSEE is split between *Engage* and *Highly-Engage*), there is every reason to expect that models transfer poorly — but without measurement this remains an assumption. A further hazard compounds the problem: on imbalanced data a model that simply predicts the majority class can post a respectable cross-dataset *accuracy* while having learned nothing transferable, so the very metric most papers report can hide the failure it should expose.

These two limitations define the gap this thesis addresses. The heterogeneity problem calls for a representation that respects the structure of the facial descriptor instead of flattening it; the generalisation problem calls for an evaluation protocol that measures transfer honestly, with metrics that cannot be gamed by majority prediction.

1.3 Research Objectives and Conceptual Framework

The objective of this thesis is to improve and, just as importantly, to *honestly characterise* four-class ordinal engagement recognition from facial-behaviour and motion features. We pursue this through one architectural idea and one evaluation discipline, organised as four research questions.

- **RQ1 — Decomposition.** Does decomposing the OpenFace descriptor into semantic groups, each with its own temporal encoder, improve engagement classification over monolithic temporal baselines that encode all 709 features together?
- **RQ2 — Encoder assignment.** If decomposition helps, which group→encoder

assignments actually matter? Is there a behavioural channel whose temporal dynamics genuinely prefer one encoder family (convolutional, recurrent, or attention-based) over another?

- **RQ3 — Motion fusion.** Does adding a global I3D motion/appearance stream to the decomposed model help, and is any gain robust across configurations rather than the product of a single lucky one?
- **RQ4 — Generalisation and losses.** How well do these models generalise across datasets, and how do loss functions trade ordinal agreement against minority-class recall and balanced ordinal error?

Conceptual framework. The unifying idea behind RQ1–RQ3, illustrated by the hybrid architecture in Chapter 3 (Figure 3.8), is to treat a clip as a *multi-stream temporal signal* rather than a single sequence. The OpenFace descriptor is split into five behaviourally meaningful groups — gaze, eye landmarks, face landmarks, head pose, and action units — and each group is encoded *independently* into a fixed-length embedding by an encoder chosen to suit its dynamics. A global I3D clip embedding can be added as a sixth stream. The embeddings are concatenated and classified, and every stream is additionally supervised by its own auxiliary head so that each behavioural channel is forced to be discriminative on its own. We call this "divide the face, then fuse" design the *semantic-group hybrid*, and it is the central object evaluated in this thesis against standard monolithic baselines, across datasets, and across loss functions. The discipline behind RQ4 is to report ordinal agreement (quadratic-weighted kappa), balanced accuracy, and balanced ordinal error (macro-MAE) as the primary metrics, never raw accuracy alone, and to test every model not only in-domain but on two unseen distributions, one of which we collected ourselves.

1.4 Contributions

This thesis makes the following contributions.

1. **A semantic-group hybrid temporal architecture** for engagement classification that decomposes the 709-dimensional OpenFace descriptor into five behavioural groups, encodes each group with an independently chosen Temporal Convolutional Network (TCN), Transformer, or LSTM, optionally fuses a global I3D motion stream, and supervises every stream with an auxiliary head (Section 3.5).
2. **A systematic 243-configuration ablation** of the per-group encoder choice, with and without the I3D stream, on CMOSE. It shows that the I3D-fused hybrid family is centred above the strongest monolithic baseline (median quadratic-

weighted kappa, QWK, of 0.553 against 0.537, best 0.605), that the benefit is a property of the family rather than of one configuration, and that head pose is the only behavioural group with a clear encoder preference, favouring a TCN — a preference that also persists under distribution shift (Section 5.2).

3. **A new, self-collected private evaluation set** of 366 webcam clips, segmented to a fixed length, filtered for tracking quality, manually labelled by the author through a purpose-built interface, and processed through the same OpenFace/I3D pipeline as the public corpora. It is used strictly as a held-out, test-only probe of real-world generalisation (Section 3.2).

4. **A 3×3 cross-dataset generalisation study** (training on CMOSE, DAiSEE, or their union; testing on each public test set plus the private set) showing that engagement models do not transfer off-the-shelf, that accuracy masks this collapse while QWK exposes it, and that pooling source corpora is the most effective simple mitigation. On the private set the architecture and the pooled-training insight *compound*: the combined-trained semantic-group hybrid is the best model (QWK 0.379 against 0.285 for the best baseline — a wider margin than in-domain), establishing the practical value of both contributions together (Sections 4.5 and 5.6).

5. **A reproducible analysis pipeline** that turns roughly 1,500 trained runs into the figures and tables of this thesis, from feature extraction through training, cross-dataset evaluation, and figure/table generation.

1.5 Organization of Thesis

The remainder of this thesis is organised as follows.

Chapter 2 reviews the literature on vision-based engagement recognition and introduces the foundational building blocks the thesis relies on. It defines the scope of the study, analyses the related feature-based and end-to-end approaches together with their strengths and weaknesses, and then presents the technical foundations: the OpenFace and I3D feature extractors, the three temporal architecture families (TCN, LSTM, Transformer), the loss functions used for imbalanced ordinal targets, and the ordinal evaluation metrics.

Chapter 3 presents the methodology. It gives an overview of the complete pipeline as a single diagram, describes the three training corpora and their class imbalance, the private test set, and the preprocessing that makes them comparable, details the monolithic baselines, and then specifies the semantic-group decomposition and the proposed hybrid architecture built on top of it. It also fixes the loss functions and their exact formulations, the training protocol, and the 3×3 cross-dataset evaluation protocol with its metrics.

Chapter 4 studies the *monolithic baselines* and fixes the experimental frame, following a protocol-first structure. An *evaluation protocol* section first fixes the three conventions every later result depends on — the primary metrics (QWK, macro-accuracy, and macro-MAE, chosen because the six metrics demonstrably disagree on the best model), CMOSE as the development corpus, and cross-entropy as the development loss (a Pareto choice among the primaries) — each justified by a-priori reasoning plus in-domain evidence, with the decisive cross-dataset confirmation deferred by explicit forward reference. Under that protocol the chapter then reads the in-domain leaderboard, analyses *prediction agreement*: how much the thirty configurations actually agree clip-by-clip, and whether the OpenFace and I3D feature families make complementary errors — the evidence that motivates fusing them in Chapter 5. Finally it takes the baselines out of domain in the 3×3 cross-dataset matrix, establishing that engagement models do not transfer off-the-shelf, and *closes the loop* on the protocol: accuracy is blind to the collapse that QWK exposes, and the loss ranking is the one in-domain preference that survives the shift.

Chapter 5 introduces the *semantic-group hybrid* and evaluates it against that frame. It first verifies that the 486-configuration hybrid population reproduces the protocol evidence of Chapter 4, then answers the design questions: which encoder suits which behavioural group (in-domain and under shift), whether fusing the I3D stream helps — a question whose answer turns out to depend on the evaluation regime — and how the best hybrid compares with the best baseline through an in-domain error analysis. The chapter then takes the hybrid out of domain: it wins every cell of the 3×3 matrix, and on the decisive self-collected private set the architecture and the corpus-pooling insight compound to the best result, showing the value of both contributions is greatest on unseen, real-world data (RQ4). Every result in both chapters is accompanied by an explicit conclusion.

Chapter 6 summarises the findings, restates the contributions in the light of the evidence, discusses the limitations of the study, and proposes concrete directions for future work, including targeting the rarest disengaged class, explicit domain adaptation, and learned rather than enumerated group encoders.

CHAPTER 2. LITERATURE REVIEW

This chapter sets the technical and scientific context for the thesis. It first fixes the scope of the study (Section 2.1), then analyses the related work and its limitations (Section 2.2). The remaining sections present the foundational knowledge the thesis builds on and that later chapters assume: the feature extractors that turn a video into numerical descriptors (Section 2.3), the three temporal architecture families used both as baselines and as the building blocks of the proposed model (Section 2.4), the loss functions for imbalanced ordinal targets (Section 2.5), and the metrics used to evaluate ordinal imbalanced classification (Section 2.6). The chapter closes by summarising how these pieces are combined in Chapter 3.

2.1 Scope of Research

This review is scoped to **vision-based automatic engagement recognition**: classifying a learner's engagement level from facial video, where the only input is a frontal webcam recording of a single person. Within this scope the thesis concentrates on four technical components, and the review is organised around them: (i) facial-behaviour and motion feature extraction, (ii) temporal sequence modelling of the per-frame descriptors, (iii) loss functions appropriate to imbalanced ordinal labels, and (iv) the metrics used to evaluate such models.

Several adjacent problems are deliberately left out of scope. We do not consider physiological-sensor engagement estimation (heart rate, electrodermal activity, eye-tracking hardware), nor interaction- or clickstream-based estimation from platform logs, because they require instrumentation beyond a webcam. We do not treat general facial-expression or emotion recognition as an end in itself, referring to it only where it informs the design of engagement features. Finally, the thesis works with *precomputed* OpenFace and I3D features as provided by the benchmark datasets; the internal training of those extractors is reviewed for understanding but is not re-implemented.

2.2 Related Work

Engagement datasets and benchmarks. DAiSEE [2] is the most widely used in-the-wild benchmark for engagement recognition. It provides short webcam clips annotated with four ordinal engagement levels and is recorded under varied, uncontrolled conditions, which makes it realistic but noisy. CMOSE [1] is a more recent, larger, expert-labelled multimodal corpus that ships precomputed OpenFace and I3D features and reports strong inter-rater reliability; its scale and label quality make it the primary dataset of this thesis. A systematic review of computer-vision engagement detection [8] surveys the common pipelines and explicitly identifies the field's two recurring difficulties — severe class imbalance and the subjectivity of engagement labels — as open problems, which is consistent with the motivation in Chapter 1.

Feature-based approaches. A large body of work first reduces each frame to a compact facial descriptor with a toolkit such as OpenFace [3] and then models the resulting sequence. Representative pipelines feed OpenFace action units, gaze, and head pose to a recurrent network, for example an OpenFace+BiLSTM model [4], or apply dimensionality reduction and resampling — principal component analysis or singular value decomposition together with synthetic minority oversampling — before a convolutional or fully-connected classifier [9]. Other studies aggregate the features over time and compare classical learners such as logistic regression, support vector machines, multilayer perceptrons, k-nearest neighbours, and gradient-boosted trees [5]. These works collectively establish that OpenFace descriptors carry a genuine engagement signal, but they share a structural assumption: the 709-dimensional descriptor is treated as a single homogeneous input vector, encoded by one model with one temporal inductive bias.

End-to-end video approaches. A second family learns directly from pixels. EfficientNetV2 [10] backbones followed by an LSTM [6] encode appearance per frame and integrate it over time, and context-aware affect-recognition models exploit spatial relationships within the scene [7]. These models can capture appearance and motion cues that landmark-based descriptors discard, but they are heavier to train, require the raw video rather than released features, and are less interpretable because the learned representation is not decomposed into nameable behavioural channels.

Gap. Two gaps persist across both families and define the contribution of this thesis. First, the *heterogeneous* facial descriptor is encoded *monolithically* rather than per behavioural subsystem, even though its constituent signals — gaze, eye and face geometry, head pose, action units — have markedly different temporal characteristics. Second, *cross-dataset generalisation is rarely measured*; in-domain numbers dominate the literature, and where accuracy is the headline metric, majority-class prediction can disguise a complete failure to transfer. The semantic-group hybrid proposed in Chapter 3 targets the first gap, and the 3×3 cross-dataset protocol of Chapter 5 targets the second.

2.3 Feature Extraction

2.3.1 OpenFace 2.0 facial-behaviour descriptors

OpenFace 2.0 [3] is an open-source facial-behaviour analysis toolkit that processes a video frame by frame and, for each frame, estimates four kinds of measurement: the two- and three-dimensional *facial landmarks* that outline the face and eye regions, the *head pose* (three translation and three rotation parameters) recovered from those landmarks, the *eye-gaze* direction vectors and angles for both eyes, and the intensity and presence of the *facial action units* of the Facial Action Coding System. Each frame is also tagged with a tracking-confidence and success flag, which downstream pipelines use to discard poorly tracked frames.

In CMOSE the per-frame descriptor used in this thesis is **709-dimensional**. A key observation, exploited throughout the thesis, is that these 709 numbers are not homogeneous: they decompose naturally into five behavioural groups by the meaning of their columns — **gaze** (8 dimensions), **eye landmarks** (280), **face landmarks** (340), **head pose** (46), and **action units** (35). These groups correspond to physically distinct subsystems with distinct dynamics: gaze and eye landmarks change rapidly and noisily; head pose drifts slowly and smoothly; action units fire in short, sparse bursts. Section 3.4 formalises this decomposition and Section 3.5 turns it into an architecture.

2.3.2 I3D motion and appearance features

The Inflated 3D ConvNet (I3D) [11] is a video representation obtained by "inflating" the two-dimensional convolutional filters of an ImageNet-pretrained image network into three dimensions, so that they operate over space and time jointly, and then pretraining the result on the large Kinetics action-recognition dataset. I3D thereby captures appearance and short-range motion that landmark-based descriptors omit. In the benchmarks used here, CMOSE and DAiSEE each provide a *single precomputed 1024-dimensional I3D embedding per clip* — a globally pooled spatiotemporal descriptor of the whole clip rather than a per-frame sequence. This thesis therefore treats I3D as a compact, clip-level motion/appearance stream that complements the per-frame OpenFace descriptors, and is careful in its claims accordingly: the I3D stream contributes a global summary, not fine-grained temporal motion.

2.4 Temporal Architectures

A clip is a sequence of per-frame OpenFace descriptors, so a model must reduce a variable-length sequence to a single engagement prediction. Three architecture families dominate sequence modelling, and all three are used in this thesis — both as monolithic baselines (Section 3.3) and as the per-group encoders inside the hybrid (Section 3.5). They differ chiefly in how they let one time step influence another.

2.4.1 Temporal Convolutional Networks (TCN)

A Temporal Convolutional Network [12] applies one-dimensional convolutions along the time axis. Two design choices make it well suited to behavioural sequences. First, the convolutions are *causal*: the output at time t depends only on inputs at times $\leq t$, which is achieved by padding on the left and removing the surplus on the right. Second, successive layers use exponentially increasing *dilation*: a layer at depth l skips 2^l steps between the inputs it convolves, so a stack of L layers reaches a receptive field that grows as $\mathcal{O}(2^L)$ while keeping the number of parameters small. Each layer is wrapped in a residual block — two dilated convolutions with weight normalisation, a rectified-linear nonlinearity, and dropout, summed with a (possibly projected) copy of the input — which stabilises the training of deep stacks. The result is a strong inductive bias for *short, local temporal motifs* at multiple scales, which matches signals such as a head nod or an action-unit burst.

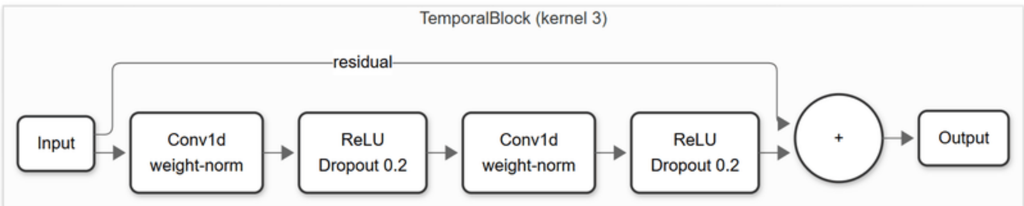


Figure 2.1: The residual TemporalBlock used in every TCN of this thesis: two weight-normalised dilated causal 1-D convolutions, each followed by a ReLU and dropout, summed with a residual connection (kernel size 3).

2.4.2 Long Short-Term Memory (LSTM)

The Long Short-Term Memory network [13] is a recurrent architecture that processes the sequence one step at a time while maintaining a cell state c_t and a hidden state h_t. Three gates — input i_t, forget f_t, and output o_t — regulate how much new information is written, how much past information is retained, and how much of the cell state is exposed:

$$\begin{aligned} f_t &= \sigma(W_f x_t + U_f h_{t-1} + b_f), & i_t &= \sigma(W_i x_t + U_i h_{t-1} + b_i), \\ o_t &= \sigma(W_o x_t + U_o h_{t-1} + b_o), & \tilde{c}_t &= \tanh(W_c x_t + U_c h_{t-1} + b_c), \\ c_t &= f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, & h_t &= o_t \odot \tanh(c_t), \end{aligned} \tag{2.1}$$

where σ is the logistic sigmoid and $\odot$ is the element-wise product. The gating mechanism lets the network carry information across many time steps without the vanishing-gradient problem of plain recurrent networks, giving it a bias toward *longer-range dependencies*. The cost is that the sequential recurrence is harder to parallelise than convolution or attention.

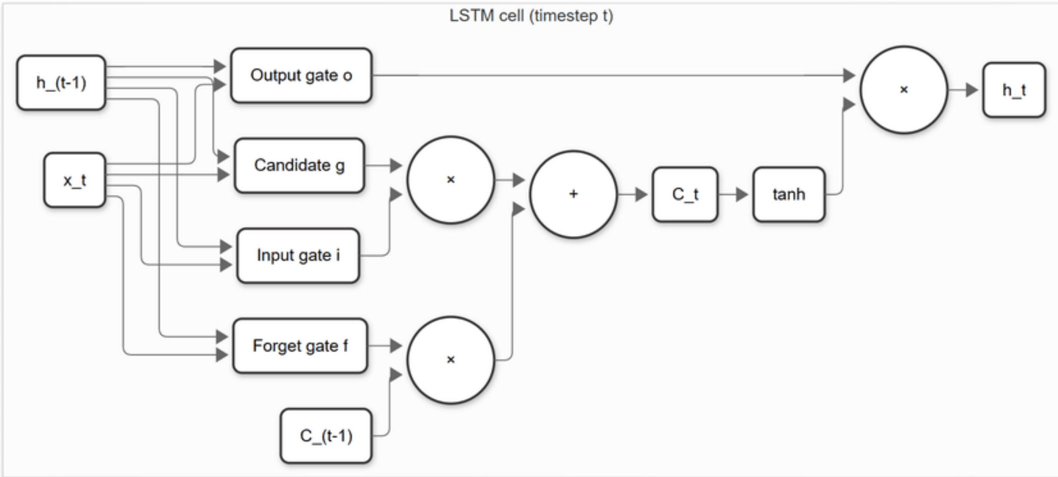


Figure 2.2: A single LSTM cell at time step t: the forget, input, and output gates regulate how the cell state C_t and hidden state h_t are updated from the previous states C_{t-1}, h_{t-1} and the input x_t.

2.4.3 Transformers

The Transformer encoder [14] dispenses with recurrence and convolution and instead relates all time steps directly through *self-attention*. For queries Q, keys K, and values V obtained by linear projections of the input, attention computes

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \tag{2.2}$$

so that every output position is a learned weighted combination of every input position, with d_k the key dimension. Because attention is permutation-invariant, order is injected through a *positional encoding* added to the input; this thesis uses the standard sinusoidal encoding. Multiple attention “heads” run in parallel and their outputs are concatenated, and each attention sub-layer is followed by a position-wise feed-forward network, with residual connections and layer normalisation throughout. The Transformer has the weakest temporal prior of the three families — it can attend anywhere — which makes it flexible but comparatively data-hungry.

2.5 Loss Functions for Imbalance and Ordinality

Let a model produce logits $z \in \mathbb{R}^K$ for the $K = 4$ engagement classes, softmax probabilities $p_k = e^{z_k}/\sum_j e^{z_j}$, and let $y \in \{0, \ldots, K-1\}$ be the ordinal ground-truth class. Three loss families are compared in this thesis; their exact application (which weights, computed on which split, combined with which auxiliary term) is given in Section 3.6.

Cross-entropy (CE) is the standard classification loss,

$$\mathcal{L}_{\text{CE}} = -\log p_y, \tag{2.3}$$

which treats all classes symmetrically and all errors as equally severe. On imbalanced data it is dominated by the majority class and on ordinal data it ignores class order, so it serves as the baseline against which the other losses are judged.

Weighted cross-entropy counteracts imbalance by scaling each class's contribution by a weight w_c inversely proportional to its frequency, $\mathcal{L}_{\text{WCE}} = -w_y \log p_y$. This raises the cost of mistakes on rare classes such as *Highly-Disengage* and pushes the model away from collapsing onto the majority, at the expense of some headline accuracy. Resampling techniques such as SMOTE [15] address the same imbalance at the data level by synthesising minority examples; this thesis instead studies loss-level remedies, which require no change to the data pipeline.

Ordinal (EMD-style) loss encodes the order of the classes by comparing *cumulative* distributions. Writing the predicted cumulative distribution as $P_k = \sum_{j \le k} p_j$ and the cumulative distribution of the one-hot label as the unit step $T_k = \mathbf{1}[k \ge y]$ (which rises from 0 to 1 at class y), the loss is the mean squared distance between the two step functions,

$$\mathcal{L}_{\text{ord}} = \frac{1}{K} \sum_{k=0}^{K-1} \big(P_k - T_k\big)^2, \tag{2.4}$$

a one-dimensional, squared Earth Mover's Distance between the predicted and target distributions over the ordered classes. Because moving probability mass across several classes costs more than moving it to an adjacent class, this loss directly penalises the large ordinal errors (HD$\leftrightarrow$HE) that accuracy and plain cross-entropy treat as ordinary.

All models are optimised with Adam [16], an adaptive first-order optimiser that maintains per-parameter running estimates of the first and second moments of the gradient and is the de-facto default for training such networks.

2.6 Evaluation Metrics

Because the labels are ordinal and imbalanced, the choice of metric is itself a research question (RQ4). This thesis reports six metrics, computed from the $K \times K$ confusion matrix C where C_{ij} counts samples of true class i predicted as j, and treats three of them as primary. Let $N = \sum_{ij} C_{ij}$.

Accuracy is the fraction of correct predictions, $\mathrm{Acc} = \frac{1}{N}\sum_i C_{ii}$. It is reported for comparability with the literature but is explicitly *not* a primary metric here, because on imbalanced data it rewards majority prediction.

Macro-accuracy (balanced accuracy) is the mean of the per-class recalls,

$$\mathrm{MacroAcc} = \frac{1}{K}\sum_{i=0}^{K-1} \frac{C_{ii}}{\sum_j C_{ij}}, \tag{2.5}$$

giving every class equal weight regardless of frequency. It exposes the minority-class collapse that accuracy hides and is a **primary** metric.

Mean absolute error (MAE) treats the class indices as an ordinal scale and averages the distance between truth and prediction, $\mathrm{MAE} = \frac{1}{N}\sum_{ij} C_{ij}\,|i-j|$, while **macro-MAE** averages the per-class MAE so that errors on rare classes are not drowned out. Both reward predictions that are at least *close* on the scale even when not exactly right; both are smaller-is-better. **Macro-MAE is a primary metric**: it is the natural ordinal-distance companion to QWK and macro-accuracy, reporting *how far* a prediction misses rather than only *whether* it misses, with each class weighted equally so the rare disengaged classes are not masked by the majority.

Cohen's kappa [17] measures agreement corrected for the agreement expected by chance. With observed agreement p_o and expected agreement p_e (computed from the marginals), $\kappa = (p_o - p_e)/(1 - p_e)$. In its weighted form it is written as $\kappa = 1 - (\sum_{ij} W_{ij}C_{ij})/(\sum_{ij} W_{ij}E_{ij})$, where E_{ij} is the expected count under independent marginals and W is a penalty matrix. The unweighted variant uses $W_{ij} = 1$ off the diagonal and 0 on it.

Quadratic-weighted kappa (QWK) is the same agreement coefficient with the quadratic penalty

$$W_{ij} = \frac{(i-j)^2}{(K-1)^2}, \tag{2.6}$$

so that disagreements are penalised in proportion to the *square* of the ordinal distance between the true and predicted classes. QWK is therefore the natural single number for this problem: it is corrected for chance, it respects the class order, and it cannot be inflated by majority prediction. It is the **primary** metric of the thesis, with macro-accuracy as the primary balanced-recall companion and macro-MAE as the primary ordinal-distance companion.

2.7 Chapter Summary

This chapter scoped the thesis to webcam-based ordinal engagement recognition and analysed the two dominant approaches in the literature, identifying their shared limitations: heterogeneous facial descriptors are encoded monolithically, and cross-dataset generalisation is seldom measured with metrics that resist majority-class inflation. It then laid the technical foundations the rest of the thesis depends on — the OpenFace 709-dimensional descriptor and its five behavioural groups, the clip-level I3D embedding, the three temporal architecture families (TCN, LSTM, Transformer) and their differing temporal biases, the three loss families for imbalance and ordinality, and the six evaluation metrics with QWK, macro-accuracy, and macro-MAE as primary. Chapter 3 combines these foundations into a concrete methodology: it turns the five-group decomposition into the proposed semantic-group hybrid architecture, specifies the baselines and the exact loss formulations, and defines the training and 3×3 cross-dataset evaluation protocols.

CHAPTER 3. METHODOLOGY

This chapter specifies the methodology of the thesis. Section 3.1 gives an overview of the complete pipeline as a single diagram. Section 3.2 describes the three training corpora, the self-collected private test set, and the preprocessing that makes them comparable. Section 3.3 defines the monolithic baseline models that anchor every comparison. Section 3.4 then formalises the semantic-group decomposition of the OpenFace descriptor, and Section 3.5 builds the proposed semantic-group hybrid architecture on top of it, in explicit contrast to those baselines. Section 3.6 fixes the exact loss formulations, Section 3.7 the training protocol, and Section 3.8 the 3×3 cross-dataset evaluation protocol and its metrics.

3.1 Overview

The task is framed as four-class ordinal classification of a fixed-length clip. Per-frame OpenFace descriptors and a clip-level I3D embedding are extracted, brought to a common temporal length, and normalised. The monolithic baselines (Section 3.3) follow the standard recipe: a single encoder consumes the full 709-dimensional OpenFace sequence (or the pooled I3D vector) and feeds a classifier. The proposed model replaces that single encoder with a split-and-encode design: the OpenFace descriptor is split into five semantic groups, each encoded independently into a 64-dimensional embedding by its own temporal encoder; in the multimodal variant the I3D clip vector is encoded into a 128-dimensional embedding by a small MLP. The embeddings are concatenated and classified by a shared multilayer perceptron, while an auxiliary head supervises every stream separately; the hybrid is shown in Figure 3.8. Every model is trained and evaluated under a 3×3 cross-dataset protocol and three loss functions.

3.2 Data and Preprocessing

3.2.1 Datasets

The thesis uses three training sources and three test sets, whose sizes and class balance are reported in Table 3.1.

- **CMOSE** [1]: 12,197 clips, divided into a `train` split (8,783), an `unlabel` split (2,193) that this thesis uses as the validation set for early stopping and checkpoint selection, and a `test` split (1,221).
- **DAiSEE** [2]: 8,571 clips, whose official `Train`/`Validation`/`Test` partitions are mapped onto the same `train`/`unlabel`/`test` roles (5,358 / 1,429 / 1,784).

- **Combined**: the union of CMOSE and DAiSEE (20,768 clips), with source-prefixed identifiers so the two corpora cannot collide. All three splits are merged split-wise, giving 14,141 / 3,622 / 3,005 train / validation / test clips.
- **Private (self-collected)**: a held-out set of **366 clips that we collected and manually labelled ourselves**, used **for testing only** — never in training or model selection. Its construction is described in Section 3.2.2.

All four sets share the four ordinal labels HD/DE/EG/HE. Table 3.1 and Figure 3.1 show that all three training corpora are heavily imbalanced toward the upper-middle classes, and — crucially — imbalanced in *different directions*. CMOSE is dominated by *Engage* (69.4%) with only 2.8% *Highly-Disengage*, whereas DAiSEE is split between *Engage* (50.3%) and *Highly-Engage* (43.8%) with just 0.7% *Highly-Disengage*. The rare disengaged classes are precisely the ones a useful monitoring system must catch, yet they amount to a fraction of a percent to a few percent of the data. The split-wise distribution (Figure 3.2) confirms that this imbalance is consistent across the train, validation, and test partitions, so it is a property of the problem rather than of a particular split.

Table 3.1: T1. Dataset statistics and class balance (%).

Dataset	Total	Train	Val	Test	HD%	DE%	EG%	HE%
CMOSE	12197	8783	2193	1221	2.800	18.100	69.400	9.600
DaiSEE	8571	5358	1429	1784	0.700	5.100	50.300	43.800
Combined	20768	14141	3622	3005	2.000	12.800	61.500	23.700

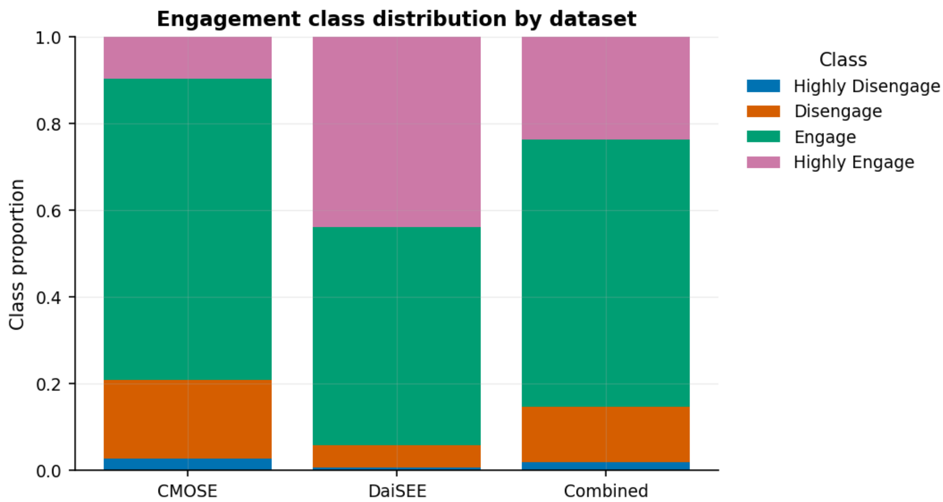


Figure 3.1: Engagement class distribution by dataset.

This severe and *heterogeneous* imbalance shapes three choices the thesis returns to. It predicts difficulty in transferring between corpora, where a model fit to

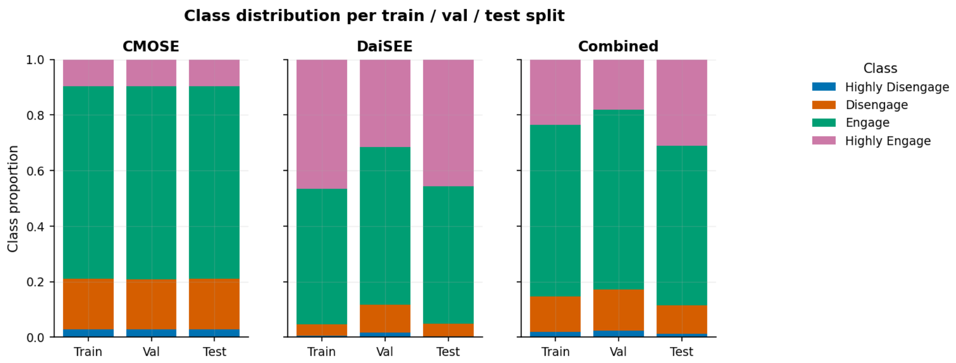


Figure 3.2: Class distribution by split, per dataset.

one label prior meets a sharply different one at test time (Chapters 4 and 5); it motivates the imbalance-aware losses of Section 3.6; and, because predicting the majority class alone scores well on every corpus, it rules out raw accuracy in favour of quadratic-weighted kappa, macro-accuracy, and macro-MAE as the **primary** metrics (Section 3.8).

3.2.2 The private evaluation set

The private set measures deployment rather than in-corpus behaviour; because no model is ever trained or selected on it, it is the cleanest probe in the thesis. It was built in five steps.

1. **Source and segmentation.** Webcam recordings of online-class sessions were sampled at several points within each session and cut into fixed-length **10-second clips**, each containing a single learner.

2. **Tracking-quality filtering.** Every clip was run through the same OpenFace pipeline as the public corpora; clips with too few successfully tracked frames were rejected automatically, since a descriptor built from unreliable frames is meaningless.

3. **Manual labelling.** The accepted clips were labelled by the author through a purpose-built local interface, one engagement label per clip; reliance on a *single annotator* is a limitation revisited in Section 6.4.

4. **Feature extraction.** Accepted clips were processed through the identical OpenFace (709-d per frame) and I3D (1024-d per clip) extraction used for CMOSE and DAiSEE, so the private features are distributionally comparable to the training features.

5. **Final set.** Of the labelled clips, the **366** with complete, aligned OpenFace and I3D features form the private evaluation set. Its label prior differs again from both public corpora — 58.2% *Engage*, 30.6% *Highly-Engage*, 8.5%

Disengage, and only 2.7% *Highly-Disengage* — which, together with its independent recording conditions, makes it the most realistic measure of how the models would behave in deployment. It anchors Section 5.6.

3.2.3 Features and preprocessing

Features. Each clip is represented by per-frame OpenFace 2.0 descriptors (709-d) [3] and a precomputed I3D embedding (1024-d) [11]. When a single OpenFace frame contains more than one detected face, the highest-confidence row is kept; non-numeric entries are coerced to zero.

Temporal resampling. OpenFace clips vary in length, so each is resampled along time to a fixed number of frames by linear interpolation per feature. The OpenFace sequence length is **300 frames** for the single-corpus runs and is reduced to **150** for the larger Combined runs to fit memory; in the multimodal variant the OpenFace stream is resampled to **75** frames. The I3D feature is a single 1024-dimensional vector per clip (not a sequence): the I3D stream mean-pools it to that vector and encodes it with an MLP (Section 3.5).

Normalisation. Every feature is standardised by a per-feature z-score whose mean and standard deviation are fit on the **training split only**; the validation, test, and private sets are transformed with those training statistics, so no target-set information leaks into preprocessing. OpenFace and I3D are standardised with separate statistics.

3.3 Baseline Models

Five monolithic baselines provide the comparison points; they form the allow-list `MODEL_ORDER` used by the analysis. Each follows the standard recipe of the literature: a *single* encoder consumes the full feature sequence and feeds a classifier.

- `openface_mlp`: flattens the OpenFace sequence and classifies it with a multilayer perceptron (→256→128→4, dropout 0.3).
- `openface_tcn`: a TCN over the 709-d sequence (residual dilated blocks of widths $256, 128, 128$; kernel 3; dilations $1, 2, 4$), adaptive average-pooling, then a $128 \to 128 \to 4$ classifier.
- `openface_lstm`: a two-layer LSTM (hidden width 256) whose final hidden state feeds a $256 \to 128 \to 4$ classifier.
- `openface_transformer`: a linear projection to 128 dimensions, a sinusoidal positional encoding, and two Transformer encoder layers (4 heads, feed-forward width 256), mean-pooled and classified.

- `i3d_mlp`: mean-pools the I3D clip feature to a 1024-d vector and classifies it with an MLP ($1024 \rightarrow 256 \rightarrow 128 \rightarrow 4$, dropout 0.3).

A sixth model, `openface_tcn_i3d_fusion` (a shared-fusion TCN over both modalities), exists in the codebase but is **excluded from the assessment** so that the comparison isolates the semantic-group hypothesis rather than generic modality fusion.

The five baseline architectures are shown in Figures 3.3–3.7. The four OpenFace baselines encode the full $T \times 709$ sequence with a single encoder, while `i3d_mlp` classifies the pooled 1024-d I3D clip vector. Forcing one encoder — and therefore one temporal bias — onto all 709 heterogeneous features at once is exactly the design decision the proposed model revisits in Sections 3.4 and 3.5.

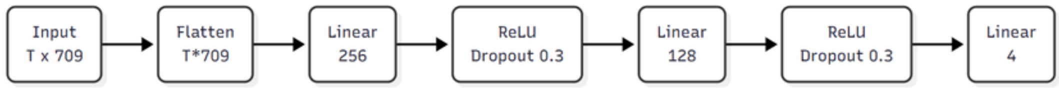


Figure 3.3: Baseline `openface_mlp`: the full $T \times 709$ OpenFace sequence is flattened and classified by a multilayer perceptron.

Figure 3.4: Baseline `openface_tcn`: a temporal convolutional network over the full 709-d sequence, adaptively average-pooled and classified.

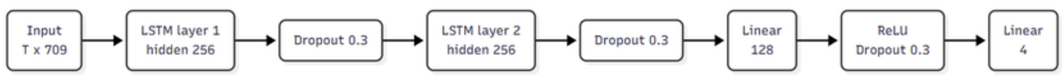


Figure 3.5: Baseline `openface_lstm`: a two-layer LSTM over the full 709-d sequence whose final state is classified.

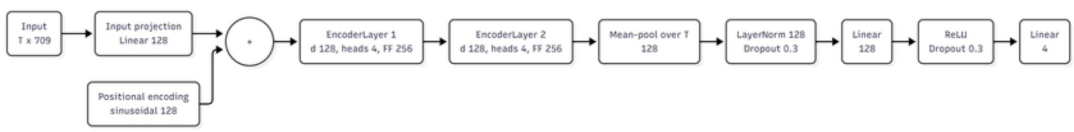


Figure 3.6: Baseline `openface_transformer`: a Transformer encoder over the full 709-d sequence with sinusoidal positional encoding, mean-pooled and classified.

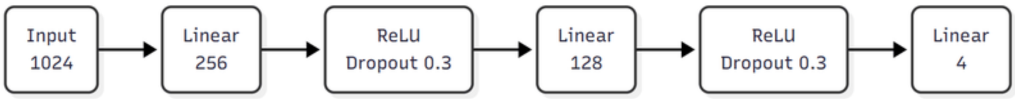


Figure 3.7: Baseline `i3d_mlp`: the pooled 1024-d I3D clip vector is classified by a multilayer perceptron.

3.4 Semantic-Group Decomposition

The 709 OpenFace features are partitioned into five behaviourally meaningful groups by matching the column names, in the fixed order used everywhere in the pipeline (gaze, eye, face, head, action units). Table 3.2 summarises the partition; every one of the 709 columns belongs to exactly one group.

Table 3.2: Semantic decomposition of the 709-dimensional OpenFace descriptor.

Group	Content	Dim
Gaze	gaze direction vectors and angles	8
Eye landmarks	2-D/3-D eye-region landmark coordinates	280
Face landmarks	2-D/3-D facial landmark coordinates	340
Head pose	head translation and rotation parameters	46
Action units	facial action-unit presence and intensity	35
Total		**709**

This decomposition is the inductive prior of the method. The five subsystems have different temporal dynamics — rapid, noisy micro-saccades in the gaze and eye groups; slow, smooth drift in head pose; brief, sparse bursts in action units — so each can in principle be matched to an encoder whose temporal bias suits it. Encoding all 709 features together, as the monolithic baselines of Section 3.3 do, forces a single bias on all of them at once.

3.5 Hybrid Architecture (Proposed)

The proposed model, shown in Figure 3.8, encodes each stream independently, fuses the resulting embeddings, and supervises every stream with its own auxiliary head. It is built from the same three encoder families as the OpenFace baselines of Section 3.3 — TCN, Transformer, and LSTM — but where a baseline applies one encoder to the full $T \times 709$ sequence, the hybrid assigns an independently chosen encoder to each of the five semantic groups of Section 3.4, so any difference in performance can be attributed to the decomposition rather than to new encoder machinery.

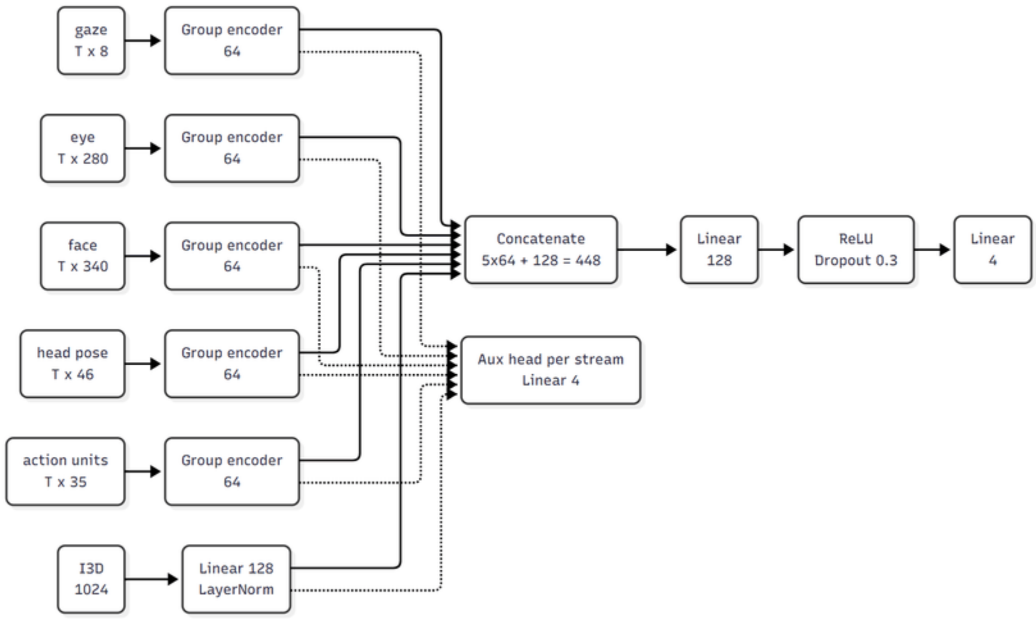


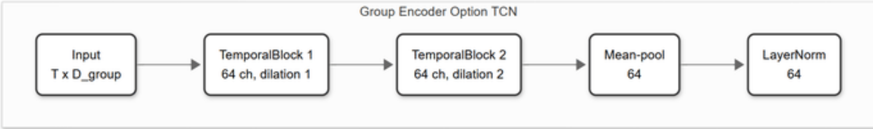
Figure 3.8: Semantic-group hybrid architecture. Each OpenFace group is encoded independently into a 64-d embedding (group encoders are chosen per group from TCN / Transformer (T) / LSTM); the I3D clip vector is encoded by an MLP into a 128-d embedding. The six embeddings are concatenated (448-d) and classified by a shared MLP; each stream also feeds an auxiliary head. The OpenFace-only variant drops the I3D row and fuses a 320-d vector.

Per-group encoder. Each group is encoded by a `GroupEncoder` that is one of three interchangeable temporal encoders (Figure 3.9), followed by temporal mean-pooling and a layer normalisation, producing a fixed **64-dimensional** embedding:

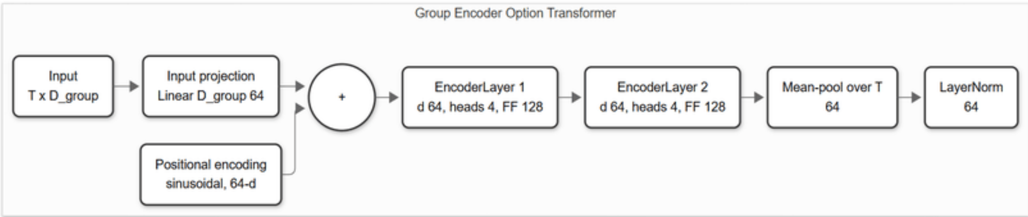
- a **TCN** — two residual blocks of dilated causal 1-D convolutions (channel widths $64, 64$; kernel size 3; dropout 0.2);
- a **Transformer** encoder — a linear projection to 64 dimensions, a sinusoidal positional encoding, and two encoder layers (4 heads, feed-forward width 128, GELU, dropout 0.2);
- an **LSTM** — a two-layer recurrent encoder of hidden width 64 (dropout 0.2).

Fusion and heads. The five 64-d group embeddings are concatenated into a 320-d vector and passed through a two-layer MLP classifier ($320 \to 128 \to 4$, ReLU, dropout 0.3) that produces the main logits. Each group additionally feeds an **auxiliary classification head** ($\mathrm{dropout} \to \mathrm{Linear}(64 \to 4)$). The total training loss is the main-head loss plus `aux_weight` $= 0.2$ times the mean of the per-group auxiliary losses (Section 3.6); the auxiliary supervision forces every stream to be individually discriminative.

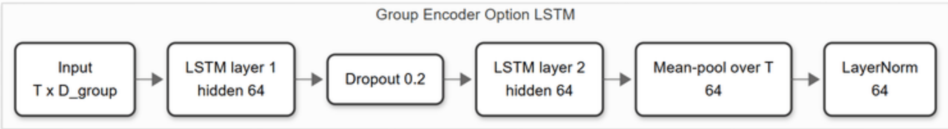
Multimodal variant. The I3D-fused model adds a sixth stream: the single 1024-



(a) TCN group encoder.



(b) Transformer group encoder.



(c) LSTM group encoder.

Figure 3.9: The three interchangeable per-group encoders. Each maps a group's $T \times D_{\text{group}}$ sequence to a 64-dimensional embedding through temporal mean-pooling and layer normalisation; one is chosen independently for each of the five groups.

d I3D clip vector is encoded by an MLP ($1024 \rightarrow 256 \rightarrow 128$, ReLU, dropout 0.3) and layer-normalised to a **128-d** embedding with its own auxiliary head. The fusion MLP then takes a **448-d** input ($5 \times 64 + 128$). The I3D stream uses an MLP rather than a temporal encoder because the provided I3D feature is a single pooled clip vector, not a sequence.

Configuration and `arch_key`. A hybrid configuration is fully specified by the encoder chosen for each of the five groups, written as an `arch_key` of five tokens in group order, for example `TCN_T_TCN_LSTM_T` (`T` = Transformer, `TCN` = TCN, `LSTM` = LSTM). With three choices per group there are $3^5 =$ **243** configurations, each evaluated with and without the I3D stream (Section 5.2).

3.6 Loss Functions

Every model is trained under each of the three losses introduced in Section 2.5: cross-entropy, weighted cross-entropy, and the ordinal (EMD) loss of Equation (2.4). The class weights for the weighted and ordinal variants are inverse-frequency weights computed on the **training split** and normalised to mean 1: for class c with n_c training samples and C non-empty classes,

$$w_c = \frac{1}{n_c} \cdot \frac{C}{\sum_k 1/n_k}, \qquad \text{so that} \quad \frac{1}{C}\sum_c w_c = 1. \tag{3.1}$$

The ordinal loss is applied in its class-weighted form as well, multiplying each sample's CDF distance by w_y.

For the hybrid models the objective also includes the auxiliary term. With main logits z and per-stream auxiliary logits $\{a^{(s)}\}$, the total loss is

$$\mathcal{L} = \mathcal{L}_{\text{main}}(z, y) \; + \; 0.2 \cdot \frac{1}{S} \sum_{s=1}^{S} \mathcal{L}_{\text{CE}}\big(a^{(s)}, y\big), \tag{3.2}$$

where $\mathcal{L}_{\text{main}}$ is the chosen loss (CE, weighted CE, or ordinal) and the auxiliary heads are always trained with plain cross-entropy, over the S streams (five groups, or six with I3D).

3.7 Training Protocol

All models are optimised with Adam [16] at a learning rate of 1×10^{-4}. Training uses mini-batches (128 for the baseline runs, 64 for the 243-configuration ablation) and **early stopping on the validation loss** — the CMOSE `unlabel` split, and its analogues for DAiSEE and Combined — with a patience of 10 epochs and an epoch cap (800 for baselines, 400 for the ablation). The checkpoint with the lowest validation loss is restored for testing. Following the convention adopted in this project, **only the training and validation loss are recorded per epoch**; no per-epoch classification metrics are computed, and the final classification metrics are evaluated exactly once, on the held-out test split. Runs use a fixed random seed (42) and optional automatic mixed precision on GPU. Representative loss curves (Figure 4.5, Chapter 4) confirm stable convergence and early stopping well before the cap, so the differences reported between models are architectural rather than optimisation artefacts.

3.8 Evaluation Protocol

Cross-dataset matrix. Generalisation is measured with a 3×3 matrix: every model and loss is trained on each of {CMOSE, DAiSEE, Combined} and evaluated on each of {CMOSE-test, DAiSEE-test, Private}. The diagonal cells are the in-domain results; the off-diagonal and the Private column measure transfer to unseen distributions.

Metrics. Each evaluation reports the six metrics defined in Section 2.6 — accuracy, macro-accuracy, MAE, macro-MAE, Cohen's kappa, and QWK — with **QWK, macro-accuracy, and macro-MAE as primary**. The empirical justification for this choice — the six metrics demonstrably disagree on the best model, and accuracy fails to detect cross-corpus collapse — is laid out in the evaluation-protocol section of Chapter 4 (Section 4.2). Row-normalised confusion matrices and per-class F1 scores are used for error analysis.

Aggregation and reproducibility. The complete study comprises roughly 1,500 trained runs: the five baselines under three losses across the three training sources, and the 243 per-group hybrid configurations (each with and without I3D) on CMOSE plus selected configurations on the other sources. All runs are aggregated programmatically, and the figures and tables of Chapters 4 and 5 are produced from the aggregate by a single command, so the entire results presentation is reproducible from the trained runs.

3.9 Chapter Summary

This chapter specified the methodology end to end: the pipeline; the three training corpora and their imbalance, the test-only private set and its collection, and the leakage-free preprocessing; the five monolithic baseline architectures (Figures 3.3–3.7); the five-group decomposition of the OpenFace descriptor; and the semantic-group hybrid (Figure 3.8) that reuses the baselines' encoder families per group, with its fusion and auxiliary heads, I3D variant, and 243-configuration design space. It also fixed the loss formulations, the training protocol, and the 3×3 evaluation protocol with its primary metrics. Chapters 4 and 5 apply this methodology to the in-domain and generalisation results respectively.

4.1 Overview

This chapter studies the *monolithic baseline* models — the standard practice of pushing a feature sequence through a single temporal encoder (Section 3.3) — and in doing so fixes the methodological frame the rest of the thesis depends on. It is organised so that every convention is justified *before* it is used. Section 4.2 fixes the **evaluation protocol**: which metrics to read, which corpus to develop on, and which loss to develop with. Each choice is argued from first principles and from evidence that is itself in-domain; the two arguments that require out-of-domain evidence — that accuracy cannot detect cross-corpus collapse, and that the loss ranking is stable under shift — are stated there as claims and proved in Section 4.5.2, so the reader is told exactly where each promise is kept. Under the fixed protocol, Section 4.3 ranks the five baselines and sets the bar the proposed architecture must clear. Section 4.4 then descends from scores to individual predictions and asks whether models with similar scores give the *same answers* — the analysis that motivates the hybrid design of Chapter 5. Section 4.5 takes the baselines out of domain with the 3$\times$3 cross-dataset matrix, closes the two deferred arguments, and asks whether in-domain strength predicts generalisation at all. Throughout, the class imbalance characterised in Section 3.2 is the backdrop: it is the reason the metrics disagree, and the reason the rare disengaged classes recur as the failure mode.

How results are aggregated. Two conventions, used in both results chapters, are fixed here. When the chapter crowns a *winner*, it takes the **maximum** over the tunable axis: the loss (and, in Chapter 5, the per-group encoder assignment) is a hyperparameter selected on validation data, so "which architecture is best" means "best achievable when tuned". Averaging over losses would punish an architecture for losses it would never ship with, and would mix objectives that deliberately optimise different metrics. When a claim concerns a *family* of configurations (Chapter 5's 243-configuration ablation, the per-group marginals), the chapter reports **means and distributions**: nobody tunes 243 configurations, so a family-level claim must hold for the population, not for its lucky maximum. Finally, every configuration is trained once with a fixed seed (Section 3.7), so wherever this thesis reports a mean it is a mean over the design grid — a measure of robustness to design choices, not a multi-seed noise estimate.

4.2 Evaluation Protocol

Before any model is compared, three conventions must be fixed. This section fixes them in order: the metrics (Section 4.2.1), the development corpus (Section 4.2.2), and the development loss (Section 4.2.3).

4.2.1 Metrics

A-priori. Engagement is an *ordinal*, severely *imbalanced* four-class problem: the levels lie on a line, and 69.4% of CMOSE is the single class *Engage* (Section 3.2, Table 3.1). Raw accuracy ignores both properties — it rewards majority-class collapse and counts a one-level miss the same as a three-level miss — whereas QWK is chance-corrected and weights every error by the squared distance between the predicted and true levels (Section 2.6). On first principles, then, accuracy is the wrong yardstick for this task and QWK the natural one. The in-domain data say the same thing twice over.

The leaderboard winner depends on the metric. Table 4.1 reports, for each of the six metrics, which of the fifteen baseline configurations (five architectures $\times$ three losses) it declares best in-domain on CMOSE. No two camps agree: QWK crowns the **OpenFace TCN under cross-entropy** (0.537); both macro metrics crown the **I3D MLP under the ordinal loss** (macro-accuracy 0.635, macro-MAE 0.458); accuracy, micro-MAE, and Cohen's kappa all crown the **I3D MLP under cross-entropy** (0.773, 0.244, 0.436). A metric choice is therefore never innocent — each crowns its own champion — so the choice must be argued, not defaulted.

Table 4.1: T8. The winning base configuration according to each metric (in-domain CMOSE).

Metric	Best model	Loss	Value
QWK	openface_tcn	CE	0.537
Macro-Accuracy	i3d_mlp	Ordinal	0.635
Macro-MAE (lower is better)	i3d_mlp	Ordinal	0.458
Accuracy	i3d_mlp	CE	0.773
MAE (lower is better)	i3d_mlp	CE	0.244
Cohen κ	i3d_mlp	CE	0.436

The six metrics form two blocks, bridged only by QWK. Figure 4.1 quantifies how far the metrics agree on the *ranking* of the fifteen configurations, as the Kendall rank correlation between every pair of metrics. The metrics split into a *micro* block — accuracy and micro-MAE, which rank the models almost identically ($\tau = 0.98$) because both are dominated by the majority class — and a *macro* block

— macro-accuracy and macro-MAE ($\tau = 0.89$) — and the two blocks are nearly orthogonal to each other (cross-block $\tau \approx 0.01$–0.10). Choosing a metric from either block alone silently discards the other half of the picture. **QWK is the only metric with moderate-to-strong agreement with both blocks** ($\tau = 0.49$–0.60, and $\tau = 0.79$ with Cohen's kappa): it is the compromise metric, sensitive to ordinal distance without being enslaved to the majority class.

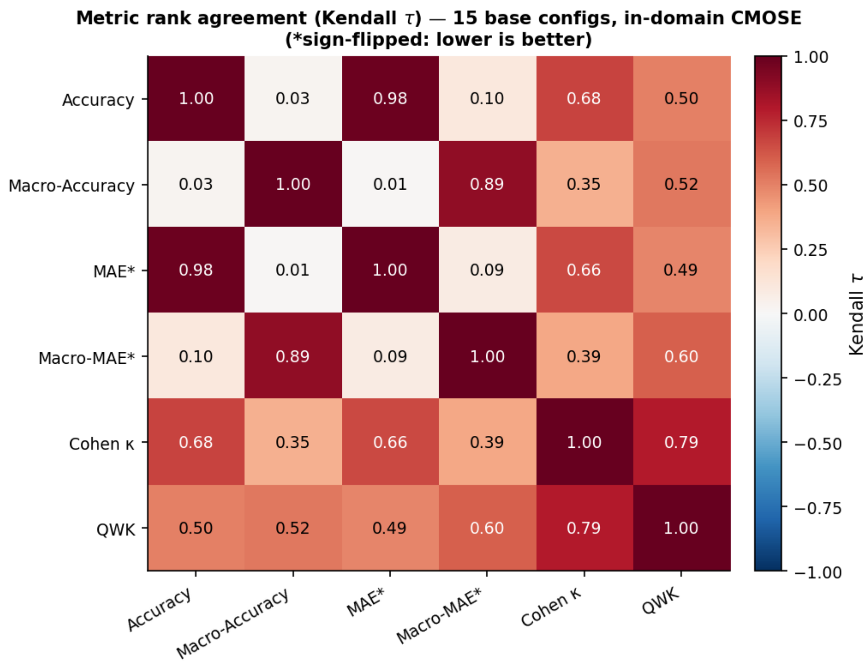


Figure 4.1: Rank agreement (Kendall τ) between the six metrics over the fifteen baseline configurations, in-domain on CMOSE (the MAE metrics are sign-flipped so that positive τ always means agreement on which models are better). The metrics form a micro block and a macro block that barely agree with each other; QWK is the only metric correlated with both.

Protocol. The thesis adopts **QWK, macro-accuracy, and macro-MAE** as the three primary metrics — an agreement coefficient, a balanced recall, and a balanced ordinal distance — with QWK as the headline; a model is judged strong only when all three agree. Accuracy and micro-MAE remain in the tables for comparability with the literature but are never used to select models. *Deferred evidence:* the decisive argument against accuracy — that it cannot even detect the collapse of a model's ordinal signal under distribution shift — requires the cross-dataset study and is delivered in Section 4.5.2.

4.2.2 Development Corpus

This choice needs no deferred evidence: it rests entirely on the two in-domain cells of the evaluation matrix. Table 4.2 and Figure 4.2 show that the best in-domain model reaches **QWK 0.537 on CMOSE but only 0.166 on DAiSEE** (macro-accuracy 0.535 versus 0.289, and macro-MAE 0.547 versus 1.226 — more than twice the ordinal error), while DAiSEE's raw accuracy of 0.548 hides the failure — a first taste of the foil that Section 4.2.1 retained accuracy to play. A corpus on which no model rises meaningfully above chance cannot separate good architectures from bad ones: whatever ranking it produces is dominated by label noise.

Table 4.2: T7. Best in-domain result per dataset (CMOSE vs DaiSEE; macro-MAE lower is better).

In-domain dataset	Best model	Loss	QWK	Macro-Acc	Macro-MAE	Accuracy
CMOSE	openface_tcn	CE	0.537	0.535	0.547	0.761
DaiSEE	i3d_mlp	CE	0.166	0.289	1.226	0.548

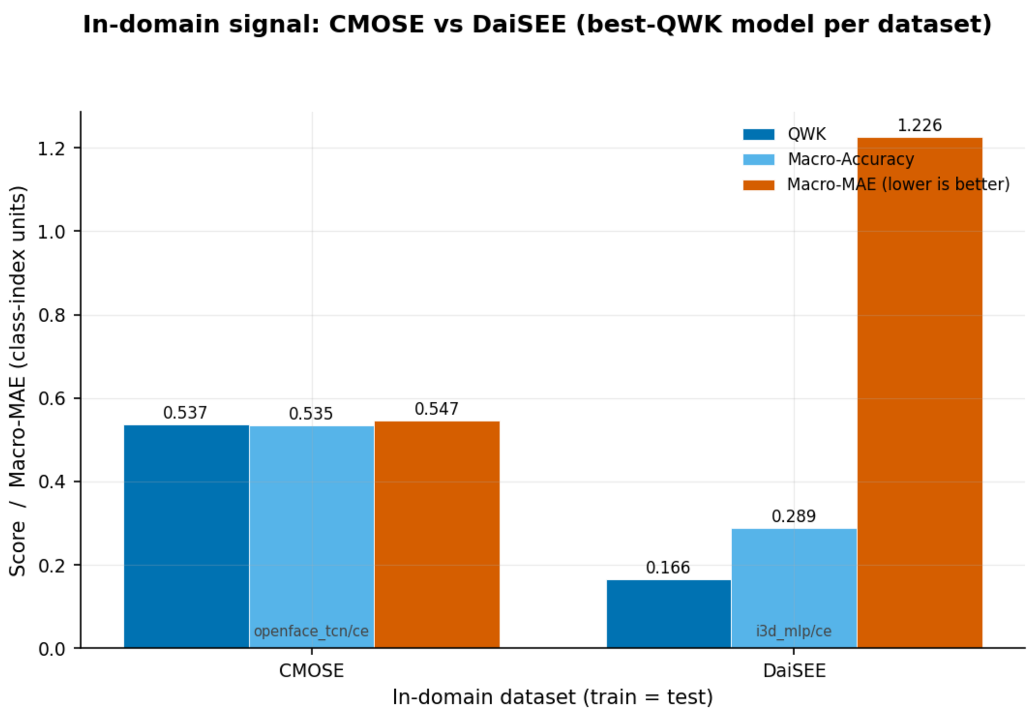


Figure 4.2: In-domain ordinal signal: CMOSE versus DAiSEE.

Protocol. CMOSE is the development corpus. DAiSEE re-enters the study as a transfer source and target in the cross-dataset matrix (Section 4.5) and in Chapter 5, where its difficulty is informative rather than obstructive.

4.2.3 Development Loss

In-domain, the loss axis is a clean trade within the primary metrics. Figure 4.3 plots, for every baseline, how each primary metric moves as the loss is rebalanced from cross-entropy through weighted cross-entropy to the ordinal loss: **macro-accuracy rises and macro-MAE drops (improves)**, while **QWK falls** for four of the five baselines. The balanced losses buy minority-class recall and a smaller balanced ordinal error at the cost of the chance-corrected agreement QWK measures.

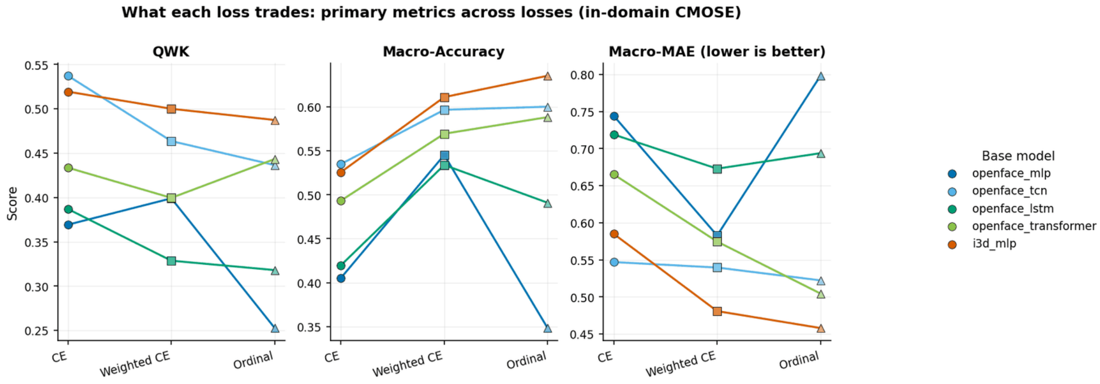
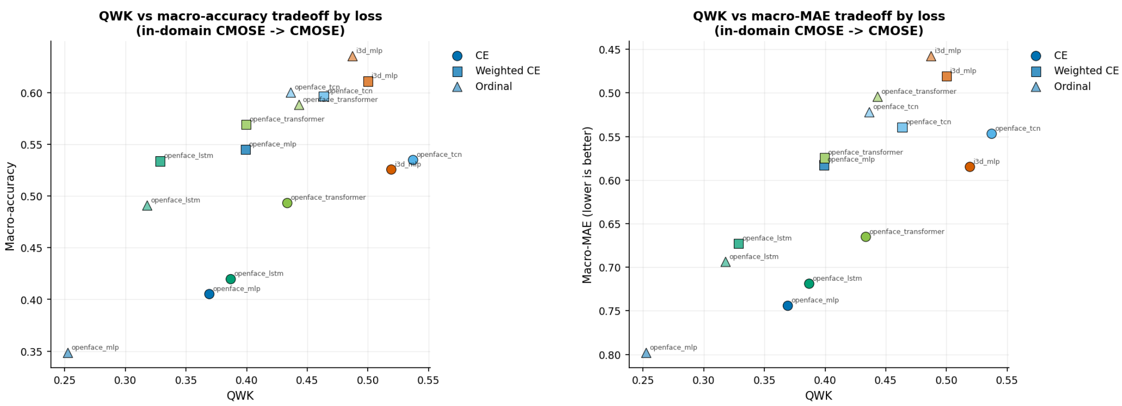


Figure 4.3: What each loss trades: the three primary metrics for every baseline across the three losses, in-domain on CMOSE. Macro-accuracy and macro-MAE improve as the loss is rebalanced, while QWK is generally highest under cross-entropy.

The same trade-off can be read as a Pareto frontier within the primary metrics themselves. Figure 4.4a plots QWK against macro-accuracy and Figure 4.4b plots QWK against macro-MAE; in both, the cross-entropy points sit toward high QWK, and the weighted-CE and ordinal points move toward better balance. For the I3D MLP, for instance, switching from cross-entropy to the ordinal loss lifts macro-accuracy from 0.526 to 0.635 and improves macro-MAE from 0.585 to 0.458 — the best balanced figures of any baseline — at the cost of QWK falling from 0.519 to 0.487. There is no loss that is best on all three primaries at once: the choice is a genuine Pareto trade-off, and for engagement monitoring — where missing disengagement is costly — the balanced losses remain a defensible operating point despite their weaker agreement.



(a) QWK vs macro-accuracy.

(b) QWK vs macro-MAE (better is up).

Figure 4.4: Two faces of the loss trade-off in the primary metrics, in-domain on CMOSE: the balanced losses move each model toward better balance on both the recall axis (macro-accuracy) and the ordinal-distance axis (macro-MAE), at the cost of QWK.

Protocol. The strongest ordinal agreement of the whole baseline grid is obtained under **cross-entropy** (best-architecture QWK by loss: CE 0.537, weighted CE 0.500, ordinal 0.487), so **cross-entropy is the development loss**: it maximises the headline primary metric and isolates the effect of the *architecture* from the confound of the loss. The semantic-group hybrid of Chapter 5 is consequently trained with cross-entropy. *Deferred evidence:* Section 4.5.2 shows that this CE > weighted CE > ordinal ranking is preserved intact under domain shift — the only loss effect that survives — which retroactively makes cross-entropy the *reliable* choice, not merely the convenient one. Figure 4.5 shows representative training and validation loss curves; all baselines converge stably and early-stop well before the epoch cap, so the differences reported in this chapter are architectural rather than optimisation artefacts.

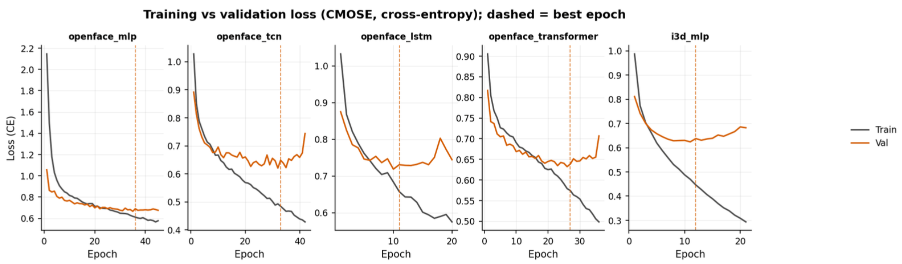


Figure 4.5: Representative training/validation loss curves (CMOSE, cross-entropy).

4.3 In-Domain Results

With the protocol fixed, the architecture question has a clean answer. Five monolithic baselines — four built on the OpenFace descriptor (an MLP, a TCN, an LSTM, and a Transformer) and one on the I3D clip vector (the I3D MLP) — are trained and tested in-domain on CMOSE under each of the three losses; Figure 4.6 and Table 4.3 report the complete grid on all six metrics. On the primary metrics, the strongest baseline is the **OpenFace TCN under cross-entropy at QWK 0.537** (max-over-loss, per the aggregation policy of Section 4.1): this is the bar the proposed architecture of Chapter 5 must clear.

The full grid adds two structural observations. First, **the loss axis decides which metric family a model wins**: cross-entropy wins the micro and agreement metrics while the rebalanced losses win the macro metrics, because only the rebalanced losses recover the rare disengaged classes that macro averaging weights equally — the per-metric winners of Table 4.1 are this pattern seen from the other side, and a direct symptom of the class imbalance of Section 3.2. Second, **the architecture axis decides the feature story**. The OpenFace TCN's fine-grained, per-frame facial dynamics carry the cues — a head turning away, gaze drifting off-screen — that

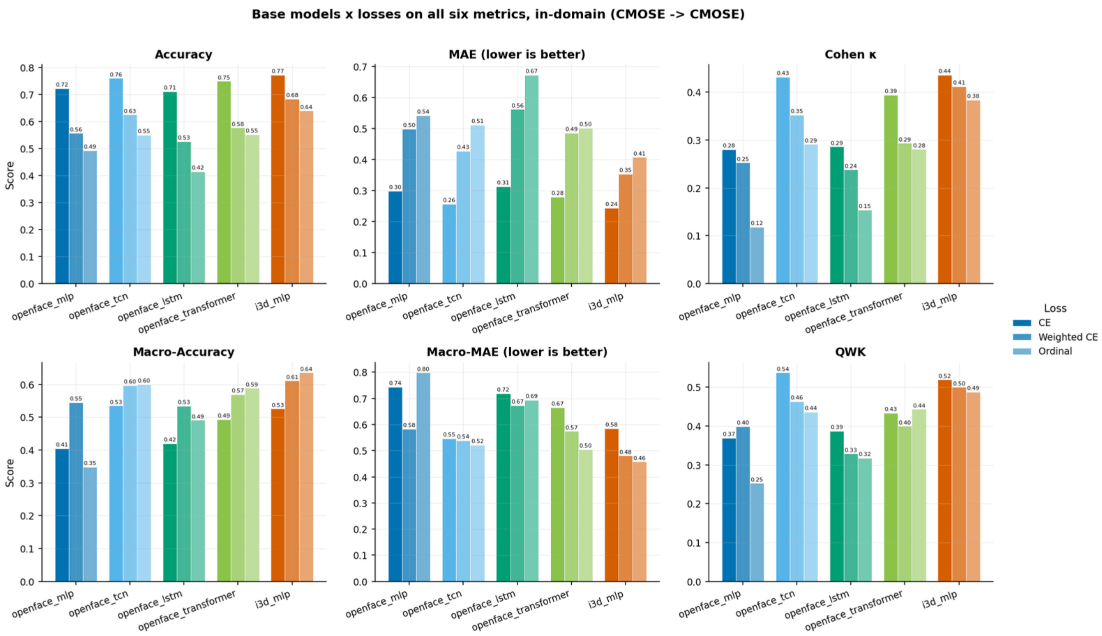


Figure 4.6: The five base models under the three losses on all six metrics, in-domain on CMOSE. The metrics disagree on the winner: accuracy, MAE, and Cohen's kappa favour the I3D MLP under cross-entropy, QWK favours the OpenFace TCN, and the macro metrics peak under the balanced losses.

separate the rarer disengaged levels and win the ordinal primary; the I3D MLP, an MLP over a single pooled appearance/motion vector, aligns with the majority *Engage* class, tops every micro metric, and is the most reweightable baseline, recovering the best balanced figures of any model under the ordinal loss. On the headline metric the two are nearly tied — **QWK 0.537 versus 0.519** — and two near-tied models built on *disjoint feature families* pose a question the score table cannot answer: are they interchangeable, or do they succeed on different clips? That question is the subject of the next section.

Table 4.3: T2. Base models × losses, in-domain (CMOSE→CMOSE), sorted by QWK.

Model	Loss	QWK	Macro-Accuracy	Macro-MAE	Accuracy	MAE	Cohen κ
openface_tcn	CE	0.537	0.535	0.547	0.761	0.256	0.432
i3d_mlp	CE	0.519	0.526	0.585	0.773	0.244	0.436
i3d_mlp	Weighted CE	0.500	0.611	0.481	0.685	0.354	0.412
i3d_mlp	Ordinal	0.487	0.635	0.458	0.640	0.408	0.383
openface_tcn	Weighted CE	0.464	0.597	0.540	0.626	0.428	0.352
openface_transformer	Ordinal	0.443	0.588	0.504	0.553	0.502	0.281
openface_tcn	Ordinal	0.436	0.600	0.522	0.551	0.512	0.291
openface_transformer	CE	0.434	0.493	0.665	0.750	0.279	0.394
openface_transformer	Weighted CE	0.400	0.569	0.575	0.577	0.486	0.293
openface_mlp	Weighted CE	0.399	0.545	0.583	0.557	0.499	0.252
openface_lstm	CE	0.387	0.420	0.719	0.712	0.314	0.286
openface_mlp	CE	0.369	0.405	0.744	0.723	0.299	0.280
openface_lstm	Weighted CE	0.329	0.534	0.673	0.527	0.563	0.238
openface_lstm	Ordinal	0.318	0.491	0.694	0.415	0.673	0.154
openface_mlp	Ordinal	0.253	0.348	0.798	0.493	0.542	0.118

4.4 Prediction Agreement

Scores summarise *how often* a model is right; they say nothing about *which clips* it is right on. This section descends to the level of individual predictions on the 1,221 in-domain CMOSE test clips and asks whether the fifteen baseline configurations give the same answers — first across the whole grid, then for the two feature families specifically. The answers motivate the central design decision of Chapter 5.

4.4.1 Agreement between Configurations

Figure 4.7 reports the pairwise Cohen's kappa between the predictions of every pair of configurations, and Table 4.4 summarises the statistics. Mean pairwise agreement is only $\kappa = 0.27$–0.39 — far below what the clustered scores of Table 4.3 suggest. Sharing the architecture ($\kappa = 0.391$) or the loss ($\kappa = 0.374$) raises agreement only mildly over sharing neither ($\kappa = 0.273$): the disagreement is driven by *both* axes, so the fifteen configurations are genuinely different hypotheses about the data, not reparametrisations of one another.

The practical consequence is **headroom**. The best single configuration classifies 77.3% of the test clips correctly, but **at least one of the fifteen is correct on 97.6%** (the oracle); only 2.4% of clips defeat every model. Yet plain majority voting over the fifteen lands at 73.5% — *below* the best single model — because the errors are too decorrelated for unweighted voting to harvest. The headroom exists, but claiming it requires a *learned* combination of complementary views of the input. That is precisely the design premise of the semantic-group hybrid of Chapter 5, which learns to fuse separately-encoded behavioural streams instead of averaging finished predictions.

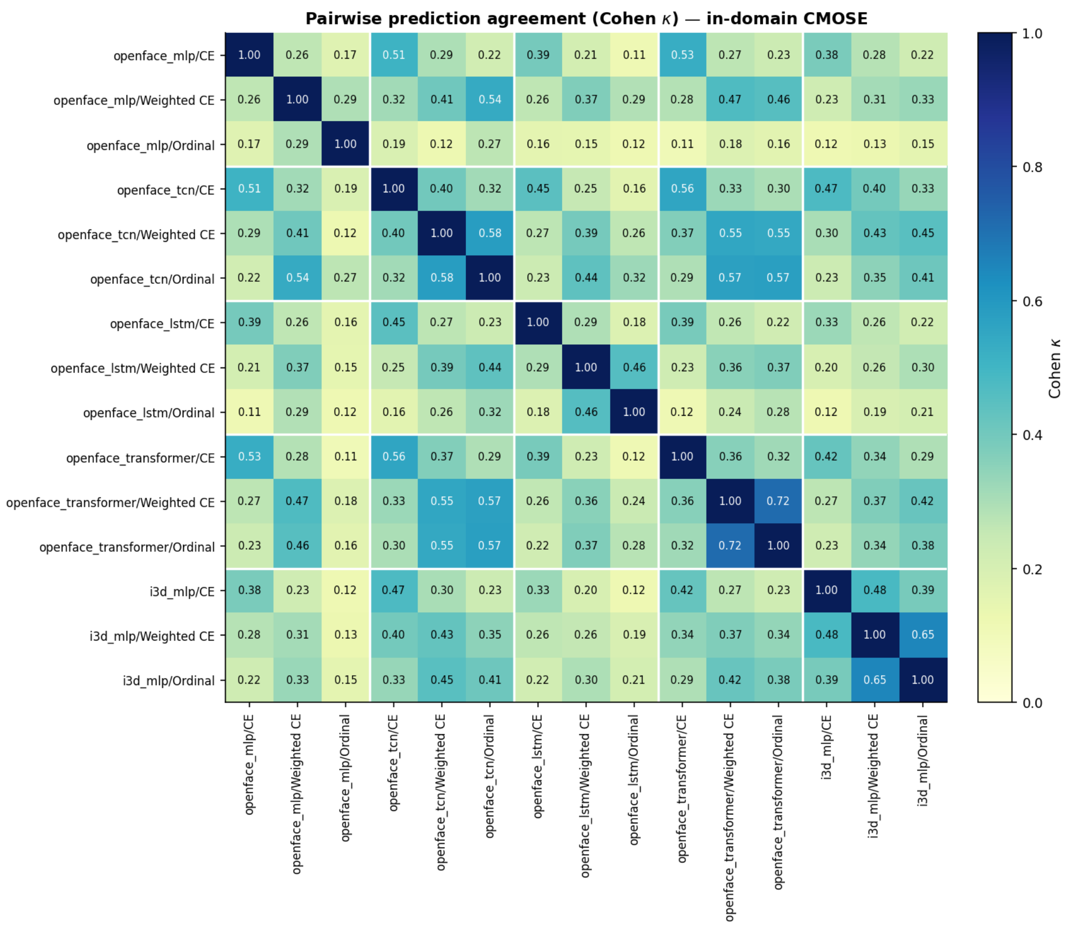


Figure 4.7: Pairwise prediction agreement (Cohen's κ) between all fifteen baseline configurations on the in-domain CMOSE test set, in model-major blocks. The diagonal blocks are the same architecture under different losses; agreement is moderate at best everywhere.

4.4.2 OpenFace versus I3D

The two feature families behind the near-tie of Section 4.3 are of very different natures: the **OpenFace** descriptor is a per-frame, structured account of facial behaviour (gaze, eye and face landmarks, head pose, action units), while the **I3D** feature is a single clip-level vector summarising global appearance and motion. Figure 4.8 compares their best models on the three primary metrics — the OpenFace TCN wins all three — but the prediction-level analysis is what reveals their relationship.

Figure 4.9 decomposes, clip by clip, where the OpenFace TCN and each other cross-entropy model are right and wrong. Against the OpenFace TCN, the **I3D MLP has the largest exclusive-correct share** — 8.6% of all test clips are clips that *only* I3D gets right, against 5–6% for the OpenFace partners — **and the smallest both-wrong share** (15.3%, against 18–19% within the OpenFace family). The pair's oracle accuracy is 84.7%, +7.4 points over the best single model and the largest pair gain in the grid (Table 4.4).

Table 4.4: T9. Prediction-level agreement and ensemble headroom of the base configurations (in-domain CMOSE).

Statistic	Value
Configs compared / test clips	15 / 1221
Best single-config accuracy	77.3%
Majority-vote accuracy (all configs)	73.5%
Oracle accuracy (>=1 config correct)	97.6%
Clips every config gets wrong	2.4%
Mean pairwise Cohen κ — same model, different loss	0.391
Mean pairwise Cohen κ — same loss, different model	0.374
Mean pairwise Cohen κ — different model and loss	0.273
openface_tcn/CE vs i3d_mlp/CE — Cohen κ	0.474
openface_tcn/CE vs i3d_mlp/CE — only openface_tcn correct	7.4%
openface_tcn/CE vs i3d_mlp/CE — only i3d_mlp correct	8.6%
openface_tcn/CE vs i3d_mlp/CE — both wrong	15.3%
openface_tcn/CE + i3d_mlp/CE — pair oracle accuracy	84.7%

Conclusion. The behaviour descriptors and the appearance features are looking at genuinely **different evidence**: the families fail on different clips. This is the empirical motivation for *fusing* the two streams in the hybrid of Chapter 5 — a motivation the score table alone could never supply, since there the I3D MLP's 0.519 looks merely redundant next to the TCN's 0.537.

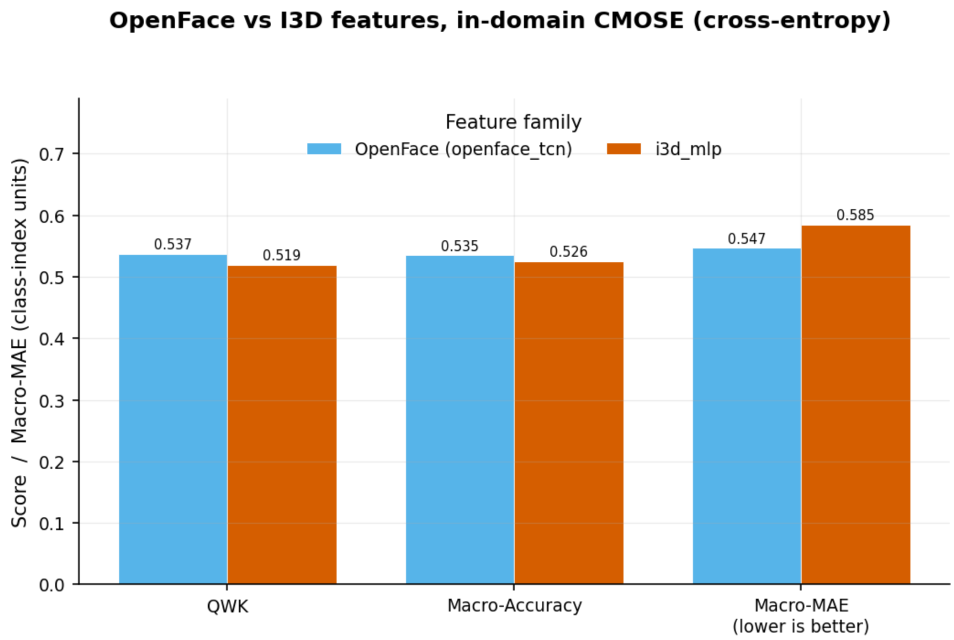


Figure 4.8: Best OpenFace encoder (the TCN) versus the I3D MLP on the three primary metrics, in-domain on CMOSE under cross-entropy: OpenFace wins all three.

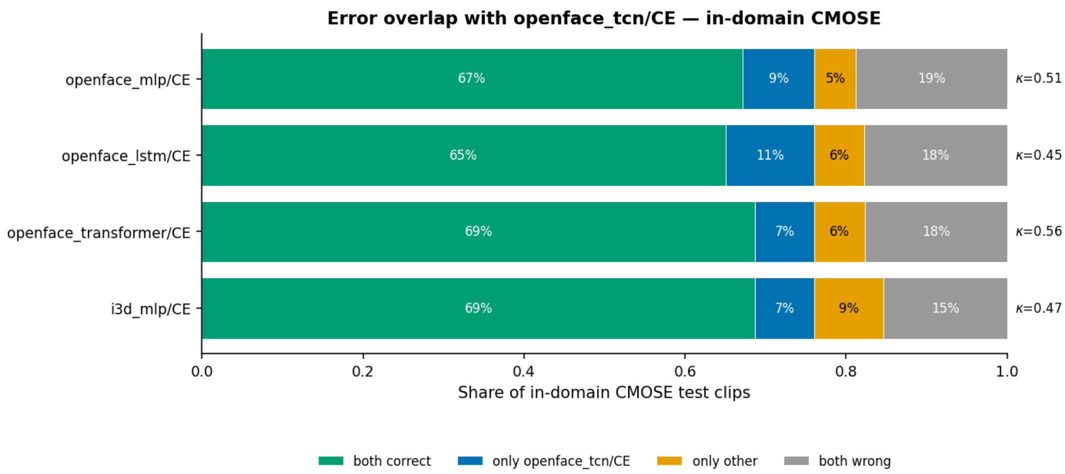


Figure 4.9: Error overlap between the OpenFace TCN and every other architecture (all under cross-entropy, in-domain CMOSE): the share of test clips both classify correctly, only one classifies correctly, or both miss. The I3D MLP has the largest exclusive-correct share and the smallest both-wrong share — the two feature families fail on different clips.

4.5 Cross-Dataset Generalization

In-domain numbers are not the deployment question. This section takes the baselines out of domain — training on one corpus and testing on another — with three aims: to measure whether any in-domain ordinal signal survives a change of corpus (Section 4.5.1); to close the two arguments the protocol of Section 4.2 deferred (Section 4.5.2); and to ask whether in-domain strength predicts generalisation at all (Section 4.5.3).

4.5.1 The 3×3 Matrix

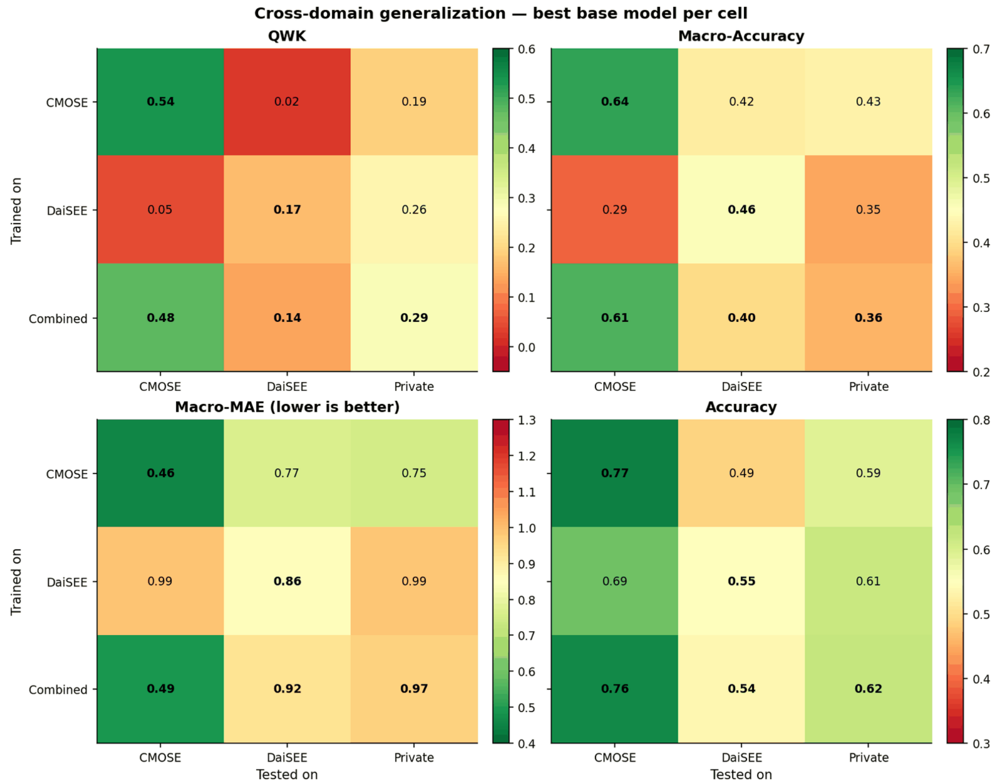


Figure 4.10: Cross-domain generalisation, best base model per cell, on the three primary metrics — QWK, macro-accuracy, and macro-MAE (lower is better) — plus raw accuracy, the deceptive foil: accuracy stays high in exactly the off-diagonal cells where every chance-corrected metric collapses.

The collapse. The 3×3 heatmaps of Figure 4.10 are the central baseline generalisation result. **On the diagonal** (train and test on the same corpus) QWK is healthy (CMOSE 0.54), macro-accuracy is well above chance (CMOSE 0.64), and ordinal error is low (CMOSE macro-MAE 0.46). **Off the diagonal all three collapse together**: CMOSE→DAiSEE QWK $= 0.02$ and DAiSEE→CMOSE $= 0.05$ (essentially chance-level agreement); macro-accuracy falls toward the 0.25 chance floor (DAiSEE→CMOSE 0.29); and macro-MAE inflates toward a full class (CMOSE→DAiSEE 0.77, DAiSEE→CMOSE 0.99) even for the best models, against ≈ 0.46 on the diagonal. The collapse is roughly symmetric — neither public corpus transfers to the other — which rules out the simpler reading that one corpus is a superset of the other and points instead to a genuine shift in both label priors and recording conditions.

The architecture ranking is reshuffled. The collapse spares no model, but it does not treat them equally. Figure 4.11 compares, per architecture, the in-domain QWK with the mean QWK over the *unseen-target* cells — the cells whose test corpus contributed no training data (the two cross-corpus cells and the private column). In-domain the order is TCN 0.537 $>$ I3D 0.519 $>$ Transformer 0.443 $>$ MLP 0.399 $>$ LSTM 0.387; on unseen targets it becomes TCN 0.112 $\approx$ Transformer 0.110 $>$ LSTM 0.096 $>$ **I3D 0.085** $>$ MLP 0.065. **The I3D MLP falls from second place to fourth**: its global appearance features bind to the corpus, while the OpenFace behaviour descriptors carry more corpus-invariant signal. In-domain performance and generalisation are therefore *complementary* selection criteria along the architecture axis, not one criterion measured twice.

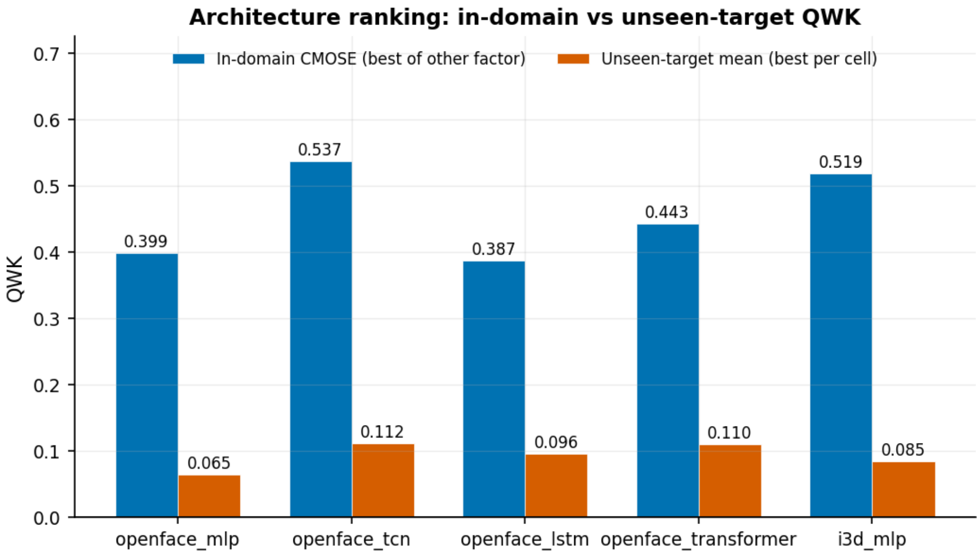


Figure 4.11: Architecture ranking in-domain versus on the unseen-target cells (best loss per cell). The in-domain order does not survive the shift: the I3D MLP drops from second to fourth, while the OpenFace encoders hold their relative positions.

The lever: pool the source corpora. The most actionable cell is the unseen **Private** column: training on the **Combined** corpus gives the best private-set QWK (0.285), beating DAiSEE-only (0.256) and CMOSE-only (0.190), while keeping near-in-domain CMOSE performance (0.477). Pooling source datasets is a simple, effective mitigation for an unknown target distribution: the union of CMOSE and DAiSEE spans a wider range of label priors than either alone, so an unseen target sits closer to the pooled distribution. This is the single most effective design lever for an unknown target, and it is carried into Chapter 5, where it is combined with the architecture contribution on the decisive private set (**RQ4**).

4.5.2 Closing the Loop: Metric and Loss Reliability

The two claims deferred from the evaluation protocol can now be proved.

Accuracy cannot see the collapse (completes Section 4.2.1). In the accuracy panel of Figure 4.10 the CMOSE→DAiSEE accuracy is a deceptively reasonable 0.49 and DAiSEE→CMOSE 0.69 — in the very cells where the QWK panel reads 0.02 and 0.05. The mechanism is the one the in-domain grid hinted at: predicting near the majority class scores well on any imbalanced target, so a model can lose every trace of transferable ordinal structure and still post a respectable accuracy by inheriting a class prior that happens to overlap the target's majority. A metric that reports business-as-usual while the model has lost all ordinal signal is not a safety net but a blindfold. Combined with its in-domain blindness to imbalance (DAiSEE's 0.548, Section 4.2.2), **accuracy is the least reliable metric in this study; QWK** — the only block-bridging metric (Section 4.2.1) and the one that detects the collapse — **is the most**. The protocol's metric choice stands confirmed, and models selected for accuracy are precisely the ones whose apparent quality evaporates under shift.

The loss ranking survives the shift (completes Section 4.2.3). Figure 4.12 repeats the per-loss comparison on the unseen-target cells: best-architecture QWK by loss is CE 0.160, weighted CE 0.099, ordinal 0.093 — the same order as in-domain (0.537, 0.500, 0.487). The loss is the only factor in the grid whose ranking transfers (the architecture ranking, by contrast, reshuffles — Section 4.5.1). Cross-entropy is therefore reliable in the strict sense that choosing it in-domain remains the right choice out of domain, which retroactively validates fixing it as the development loss.

4.5.3 In-Domain versus Transfer

A final question completes the generalisation picture: *does in-domain strength predict transfer?* Figure 4.13 plots, for every trained model (each configuration trained on CMOSE or on DAiSEE), its in-domain QWK against its mean QWK over its unseen-target cells. The Spearman rank correlation over the thirty points is only $\rho = 0.28$: a link too weak to select models on. In-domain rank is a poor compass for deployment — model selection for an unknown target cannot read the in-domain leaderboard alone and must lean on the pooling lever of Section 4.5.1. Chapter 5 sharpens this warning on the hybrid population, where the correlation even turns negative, and shows that the proposed architecture improves transfer by *lifting* the whole trade-off rather than by riding it.

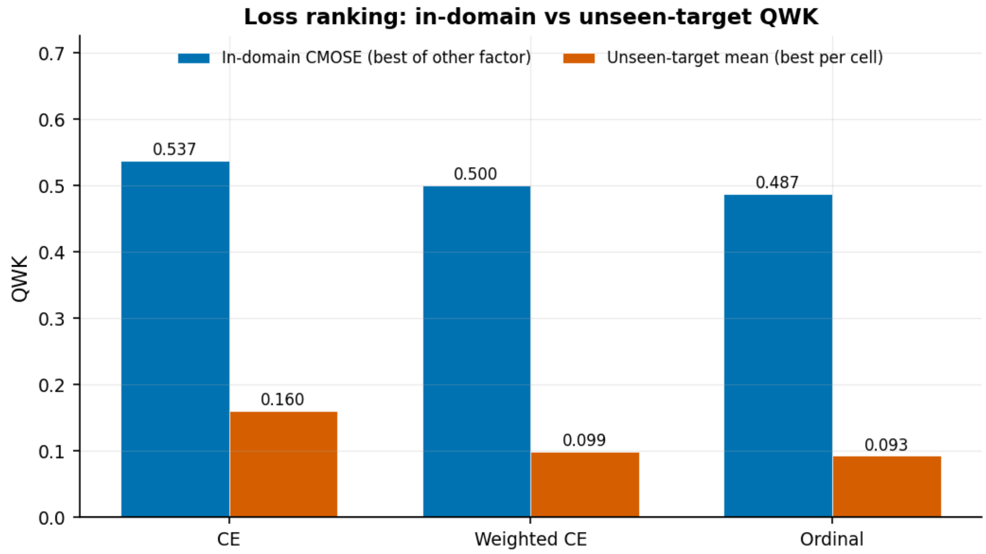


Figure 4.12: Loss ranking in-domain versus on the unseen-target cells (best architecture per cell): cross-entropy is best in both regimes — the only factor ranking that survives the shift.

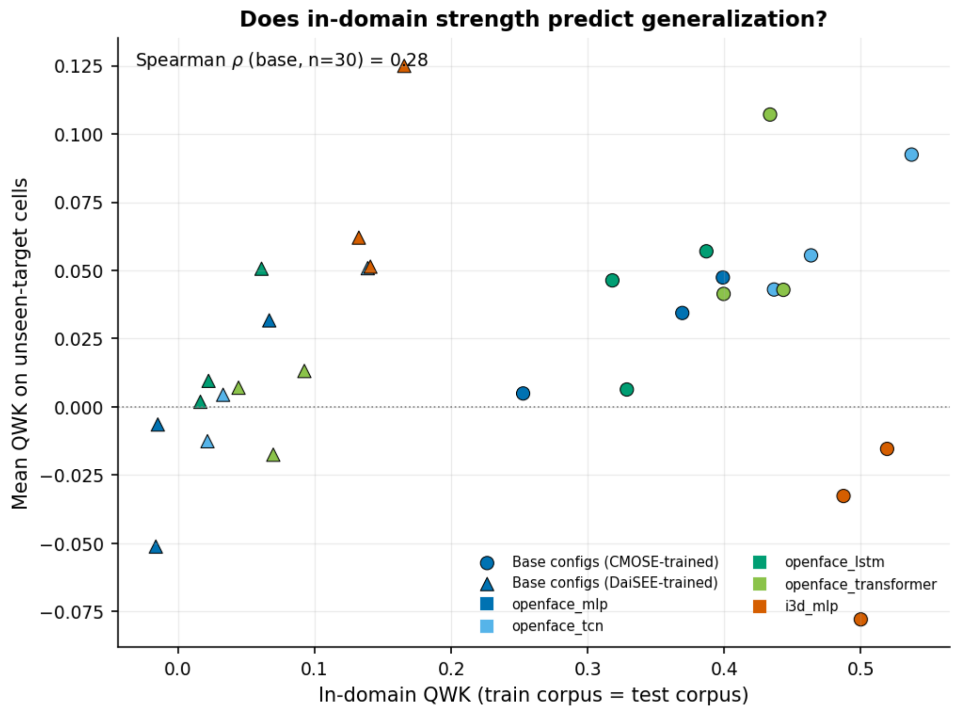


Figure 4.13: In-domain QWK versus mean QWK on the unseen-target cells, one point per trained baseline configuration (circles: CMOSE-trained; triangles: DAiSEE-trained; colour: architecture). The correlation is weak ($\rho = 0.28$): in-domain strength barely predicts transfer.

4.6 Chapter Summary

This chapter fixed the experimental frame — protocol first, results after — and characterised the monolithic baselines. The evaluation protocol (Section 4.2) fixed **QWK, macro-accuracy, and macro-MAE** as the primary metrics — each of the six metrics crowns a different winner, and QWK is the only metric bridging the otherwise orthogonal micro and macro blocks — **CMOSE** as the development corpus (in-domain QWK 0.537 against DAiSEE's 0.166), and **cross-entropy** as the development loss (the QWK-maximising end of a genuine Pareto trade-off). Under that protocol the strongest baseline is the OpenFace TCN at **QWK 0.537**, the bar the hybrid must beat (Section 4.3). The prediction-level analysis (Section 4.4) showed that the fifteen configurations agree far less than their scores suggest ($\kappa = 0.27$–0.39), that an oracle over them would reach 97.6% against the best single model's 77.3% — headroom that naive voting cannot claim — and that the OpenFace and I3D families fail on *different* clips, the direct motivation for fusing them. The cross-dataset study (Section 4.5) showed that baselines do not transfer off-the-shelf — off-diagonal QWK collapses to near zero and the architecture ranking reshuffles, with I3D falling furthest — and closed the two deferred arguments: accuracy stays deceptively high in exactly the cells where the ordinal signal disappears, and the loss ranking is the only one that survives the shift, vindicating both protocol choices; corpus pooling emerged as the most effective simple mitigation, and in-domain rank proved a weak predictor of transfer ($\rho = 0.28$). Chapter 5 now introduces the semantic-group hybrid and, under exactly this frame, asks whether learning to fuse the complementary streams beats these baselines in-domain and whether the gain compounds with pooling on unseen, real-world data.

5.1 Overview

Chapter 4 fixed the evaluation protocol — QWK, macro-accuracy, and macro-MAE as the primary metrics; CMOSE as the development corpus; cross-entropy as the development loss; the OpenFace TCN at QWK 0.537 as the bar — and supplied the design motivation: the baseline configurations err on different clips, leaving headroom that naive voting cannot claim, and the OpenFace and I3D feature families in particular fail on different clips. This chapter evaluates the proposed **semantic-group hybrid** (specified in Section 3.5) against that frame: all $3^5 = 243$ per-group encoder configurations, each with and without the I3D stream, trained with cross-entropy. Section 5.2 first re-validates the Chapter 4 frame on this 486-configuration population — if the protocol were an artefact of five hand-picked baselines, that is where it would break (RQ1 in part). Section 5.3 isolates which behavioural group's encoder choice matters, in-domain *and* under shift (RQ2). Section 5.4 measures the effect of I3D fusion with a paired ablation across evaluation regimes (RQ3). Section 5.5 pins down the headline in-domain comparison with the baselines and localises the gain class by class (RQ1). Section 5.6 then takes the hybrid out of domain: it confirms that the cross-dataset collapse is architecture-independent, and establishes the new result that the hybrid generalizes better than the baselines — in *every* cell of the matrix, and most of all on the decisive self-collected private set, where the architecture and the corpus-pooling lever *compound* (RQ4). Section 5.7 discusses why, and reads the headline number honestly. As in Chapter 4, accuracy appears only as a foil; every conclusion is argued from the primary metrics, with family-level claims reported as means and distributions and headline comparisons as the tuned maximum (the aggregation policy of Section 4.1).

5.2 Consistency with the Baseline Frame

The ablation population doubles as a stress test of Chapter 4's protocol: 486 configurations of a different architecture family, none of which existed when the frame was fixed. Figure 5.1 plots the distribution of the two hybrid families (with and without I3D) on every metric, each panel against the best base model on that metric. Three findings of Chapter 4 reproduce.

The metric structure reproduces. Recomputing the Kendall rank correlation between the six metrics over the 486 in-domain configurations (Figure 5.2) yields the same two-block picture as Section 4.2.1: a micro block ($\tau = 0.87$), a macro block ($\tau = 0.73$), weak cross-block agreement ($\tau = 0.26$–0.33), and QWK again the

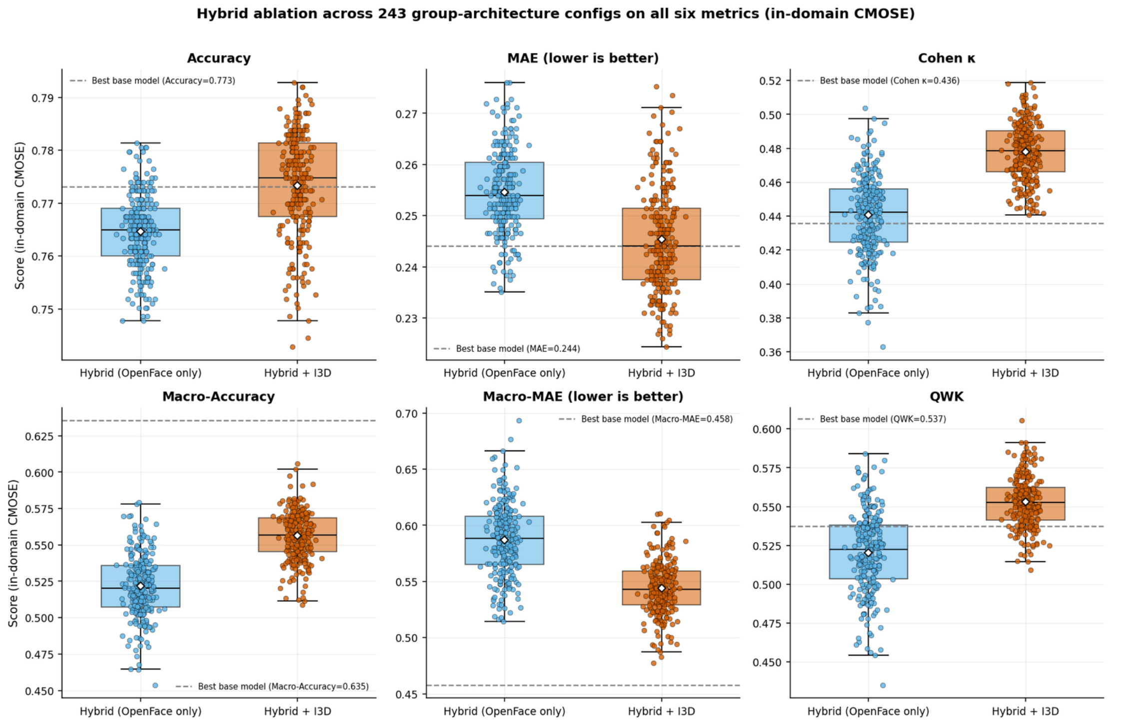


Figure 5.1: Hybrid ablation across all 243 per-group encoder configurations, with and without I3D, on all six metrics (in-domain CMOSE). Each dashed line is the best base model on that metric, across all three losses.

only bridge ($\tau = 0.63$–0.73 with both blocks). The block structure is a property of the metrics on this task, not of any model family.

Accuracy is still blind. In Figure 5.1 every configuration sits in a narrow accuracy band around the best baseline's 0.773 (top-left panel), so a model chosen on accuracy would see no architecture signal at all, while the QWK panel separates the families cleanly — the same blindness Section 4.2.1 argued for the baselines, now at the scale of 486 configurations. The macro panels carry one honest caveat: their dashed lines (macro-accuracy 0.635, macro-MAE 0.458) belong to the *ordinal-loss* I3D MLP, a balanced-loss operating point that buys those figures by giving up QWK (0.487, below the bar) — the loss trade-off of Section 4.2.3, not an architecture effect; no CE-trained model, baseline or hybrid, reaches it.

The cross-domain collapse reproduces. Taken out of domain (Section 5.6.1), the hybrid matrix shows the same shape as the baseline matrix: a healthy diagonal (0.605 on CMOSE, 0.228 on DAiSEE) and an off-diagonal collapse, with accuracy again masking it. The collapse is therefore a property of the *problem* — divergent label priors and recording conditions — not of a particular architecture.

Conclusion. All three pillars of the Chapter 4 frame survive a 32-fold larger model population. The protocol carries over without amendment, and the findings it produced can be treated as task properties (RQ1's frame condition); the remaining

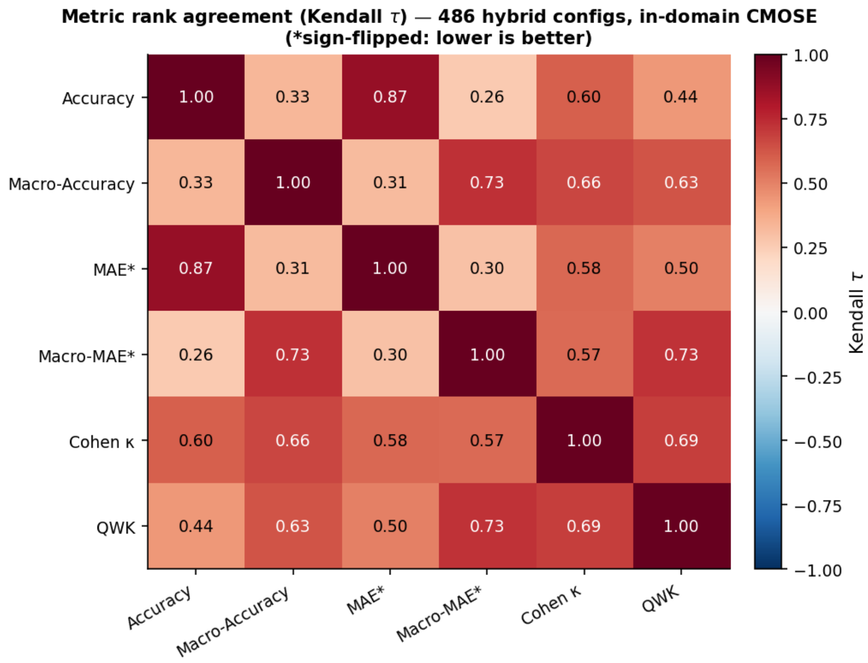


Figure 5.2: Rank agreement (Kendall τ) between the six metrics over the 486 hybrid configurations, in-domain on CMOSE: the same two-block structure as for the baselines (Figure 4.1), with QWK again the only bridging metric.

sections use it to answer the architecture questions.

5.3 Encoder Choice per Group

The 243-configuration grid varies five group→encoder choices at once; this section marginalises them to find which group's choice actually moves the primary metrics (RQ2) — and, new to this analysis, whether any preference it finds is corpus-specific or survives domain shift.

In-domain. Table 5.1 and Figure 5.3 show the mean QWK when each group uses a TCN, a Transformer, or an LSTM, marginalising over all other groups. Only **head pose** shows a clear preference — TCN (0.545), ahead of LSTM (0.541) and Transformer (0.523), a spread of 0.022. Macro-MAE agrees: head pose is again the only group with a meaningful spread, lowest (best) under the TCN (0.559) and worst under the Transformer (0.574). The other four groups are essentially encoder-agnostic (QWK spread ≤ 0.006).

Table 5.1: T5. Per-group marginal effect of encoder choice on in-domain QWK.

Group	Mean QWK (TCN)	Mean QWK (Transformer)	Mean QWK (LSTM)	Spread	Prefers
Gaze	0.539	0.537	0.534	0.006	TCN
Eye landmarks	0.538	0.537	0.535	0.003	TCN
Face landmarks	0.537	0.534	0.539	0.006	LSTM
Head pose	0.545	0.523	0.541	0.022	TCN
Action units	0.536	0.537	0.537	0.001	Transformer

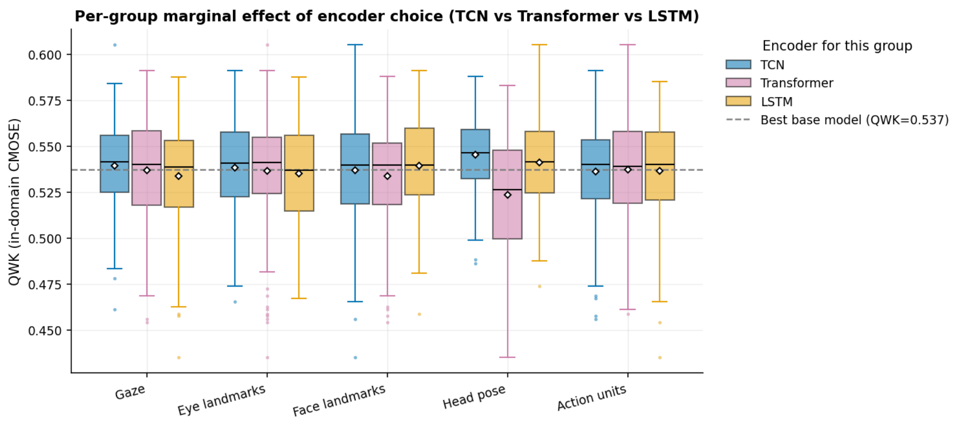


Figure 5.3: Per-group marginal effect of the encoder choice on in-domain QWK.

Under shift. Table 5.2 and Figure 5.4 recompute the same marginal over the unseen-target cells. Head pose again has the largest spread (0.010) and again prefers the TCN; the other groups' spreads (0.002–0.004) flip winners arbitrarily — noise. The consistency across regimes is what elevates the head-pose rule from a tuning accident to a property of the signal: head dynamics such as nodding and turning away are temporally local and convolution-friendly, while the gaze, landmark, and action-unit streams are insensitive to the encoder choice in both regimes.

Table 5.2: T10. Per-group marginal effect of encoder choice on QWK, pooled over the unseen-target cells.

Group	Mean QWK (TCN)	Mean QWK (Transformer)	Mean QWK (LSTM)	Spread	Prefers
Gaze	0.093	0.095	0.094	0.002	Transformer
Eye landmarks	0.096	0.092	0.094	0.004	TCN
Face landmarks	0.092	0.096	0.093	0.004	Transformer
Head pose	0.100	0.091	0.090	0.010	TCN
Action units	0.093	0.096	0.093	0.003	Transformer

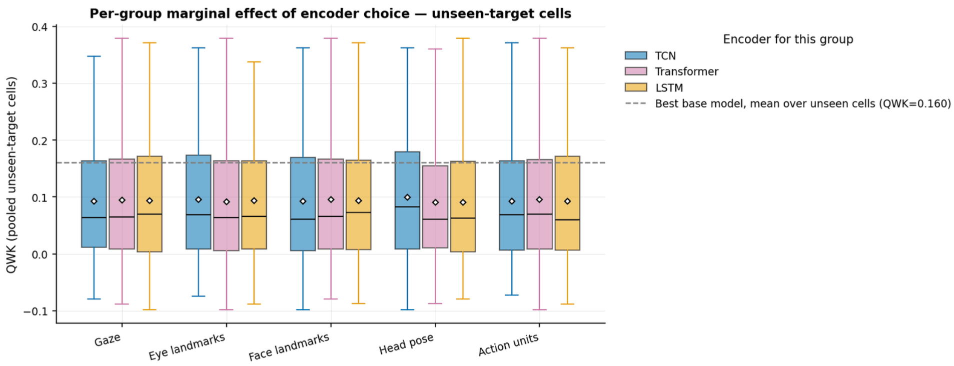


Figure 5.4: The same per-group marginal computed over the unseen-target cells: head pose again shows the largest spread and again prefers the TCN, so the in-domain preference is not corpus-specific.

Conclusion. Of the five behavioural groups, only head pose has a clear encoder preference — a TCN — and that preference survives domain shift (RQ2). The decomposition is a sound, interpretable framework whose tunable degrees of freedom are fewer than the 243-cell grid suggests, and most of its benefit must therefore come from *decomposing and fusing* rather than from the per-group encoder search — which is exactly what the next section measures.

5.4 Effect of I3D Fusion

Chapter 4 predicted, from the error-overlap analysis of Section 4.4.2, that fusing the I3D stream should pay. This section measures the prediction with the strongest design available in the grid: **pairing each OpenFace-only configuration with its exact +I3D twin** (same five encoders), so that the distribution of paired QWK differences *is* the I3D effect, free of the configuration as a confounder. Figure 5.5 reports the paired differences in three evaluation regimes: the cells whose test corpus was seen in training, the two cross-corpus cells, and the private set.

Where the test corpus is seen, fusion pays decisively. Over the seen-target cells the mean paired gain is **+0.060 QWK and 94% of the 972 pairs improve**. In-domain on CMOSE this is the cash value of the complementarity measured in Section 4.4.2: the I3D-fused family is centred above the OpenFace-only family (median QWK 0.553 against 0.522) and above the baseline bar — **82% of the**

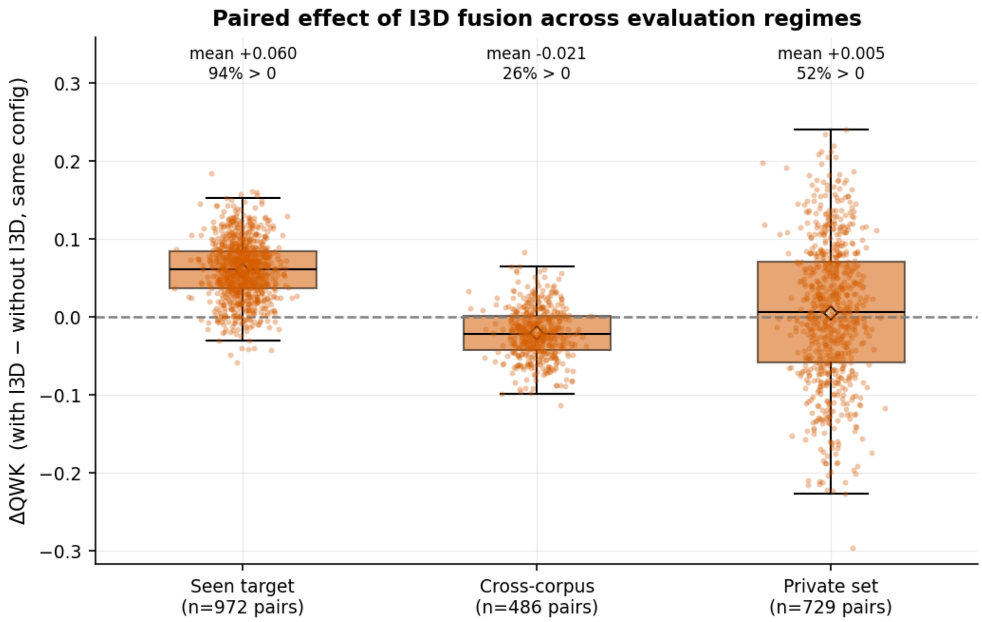


Figure 5.5: Paired effect of I3D fusion: ΔQWK between each configuration with and without the I3D stream (same per-group encoders), by evaluation regime. Fusion is decisively positive where the test corpus was seen in training, mildly negative cross-corpus, and neutral with high variance on the private set.

243 fused configurations (199) clear 0.537, against 26% of the OpenFace-only ones — with the like-for-like balanced metrics improving as well (fused medians: macro-accuracy 0.557, macro-MAE 0.543, against the best CE baseline's 0.535 and 0.547) and Cohen's kappa telling the same story (fused median 0.479 against the best base 0.436). The gain is a property of the *family*, not of one lucky configuration (RQ3).

Beyond the training corpora, the gain evaporates. Cross-corpus the mean paired effect is -0.021 **QWK with only 26% of pairs improving**; on the private set it is +0.005 with 52% improving and very high variance (± 0.2). I3D's global appearance features drag corpus bias along — the same mechanism behind the I3D MLP's fall from second to fourth place in the architecture ranking under shift (Section 4.5.1); the baseline result and the ablation result are one finding seen at two scales.

Conclusion. Adding the I3D stream is decisively beneficial when the deployment domain is represented in training, and neutral-to-harmful beyond it (RQ3). The practical rule: *fuse I3D when the target domain is in the training distribution; rely on the OpenFace groups otherwise.* Note the rule concerns the family average — the single best private-set model in Section 5.6.2 does carry I3D, but it is one tail of a wide, zero-centred distribution.

5.5 Comparison with Baselines

This section pins down the headline in-domain comparison on the primary metrics and then opens it up class by class to see where the gain actually lands.

Best hybrid versus best baseline. Table 5.3 lists the top-10 configurations; the best overall is the I3D-fused hybrid `TCN_T_TCN_LSTM_T`, and every top configuration mixes at least two encoder types. Figure 5.6 summarises the headline comparison on the three primary metrics: the best hybrid improves on the best baseline by **+0.068 QWK** (0.605 versus 0.537), lifts macro-accuracy (0.582 versus 0.535), and lowers macro-MAE (0.489 versus 0.547, lower is better). All three primary metrics move in the same direction, so the gain is real — but **modest**.

Table 5.3: T4. Top-10 hybrid configs in-domain (CMOSE), sorted by QWK (macro-MAE lower is better).

Variant	Arch (gaze_eye_face_head_au)	QWK	Macro-Acc	Macro-MAE	Accuracy
Hybrid + I3D	TCN_T_TCN_LSTM_T	0.605	0.582	0.489	0.784
Hybrid + I3D	T_TCN_TCN_LSTM_T	0.591	0.606	0.483	0.787
Hybrid + I3D	T_T_LSTM_LSTM_TCN	0.591	0.563	0.505	0.781
Hybrid + I3D	T_TCN_T_TCN_T	0.588	0.566	0.531	0.789
Hybrid + I3D	LSTM_LSTM_TCN_LSTM_T	0.588	0.567	0.517	0.787
Hybrid + I3D	T_T_LSTM_LSTM_LSTM	0.585	0.578	0.507	0.788
Hybrid + I3D	T_TCN_LSTM_TCN_T	0.585	0.573	0.514	0.779
Hybrid + I3D	LSTM_LSTM_LSTM_LSTM_T	0.585	0.580	0.494	0.773
Hybrid + I3D	T_TCN_LSTM_LSTM_TCN	0.584	0.569	0.527	0.784
Hybrid (OpenFace only)	TCN_LSTM_TCN_TCN_LSTM	0.584	0.578	0.518	0.781

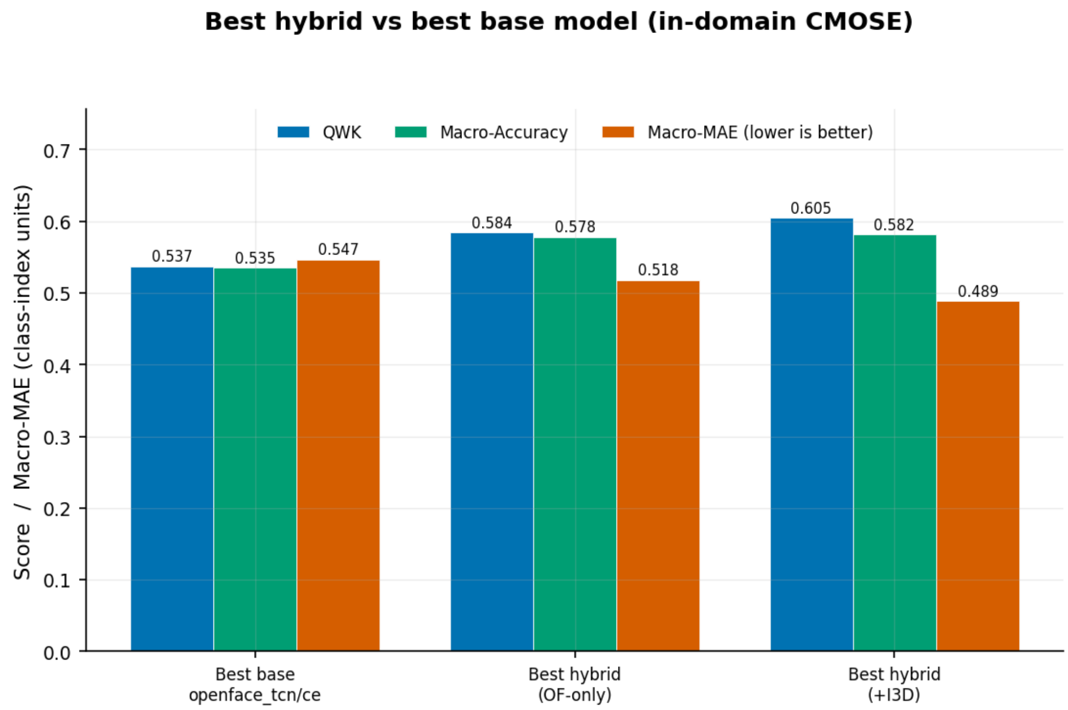


Figure 5.6: Best hybrid versus best base model, in-domain on CMOSE.

Where does the gain come from? The headline QWK gain is an average; the row-normalised confusion matrices (Figure 5.7) and the per-class F1 scores (Figure 5.8) localise where the hybrid helps and where the problem remains open.

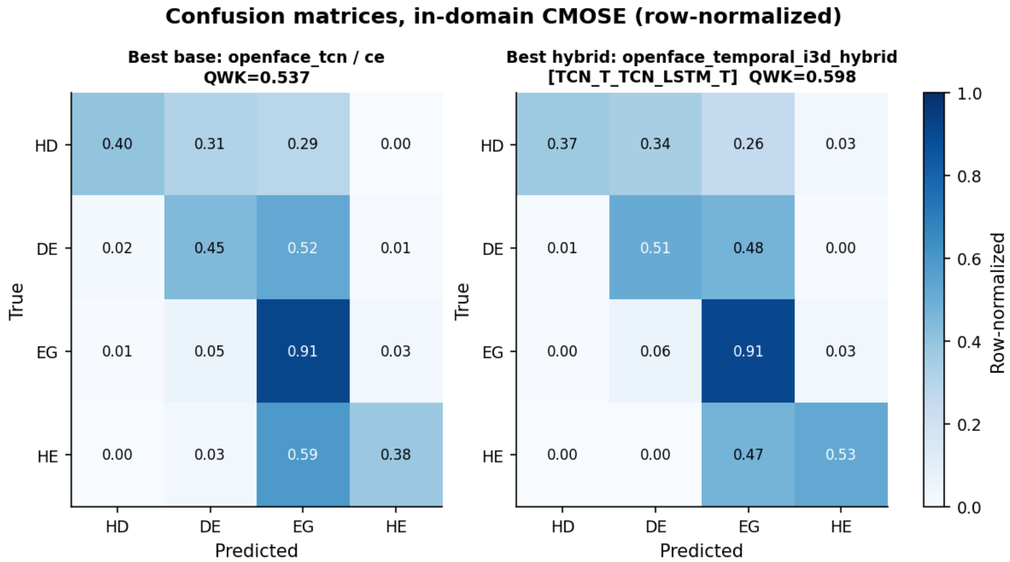


Figure 5.7: Confusion matrices of the best base versus the best hybrid (in-domain, row-normalised).

Relative to the best baseline, the hybrid improves recall on the ordinal extreme **Highly-Engage** ($0.38 \to 0.51$) and on **Disengage** ($0.45 \to 0.55$), while keeping *Engage* near 0.90 ($0.91 \to 0.90$). However, the rarest class **Highly-Disengage** stays unsolved and is slightly traded away (base recall $0.40 \to$ hybrid 0.37): the model reallocates capacity toward the more frequent extreme.

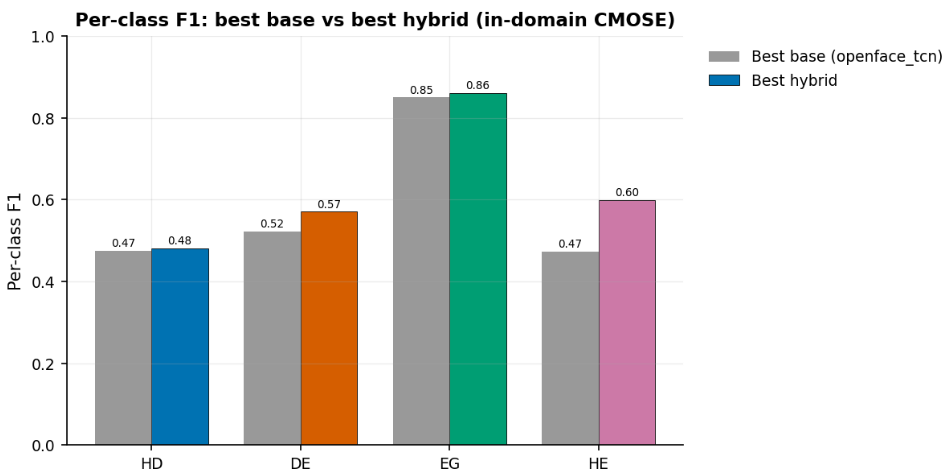


Figure 5.8: Per-class F1: best base versus best hybrid.

Conclusion. The hybrid's in-domain improvement is real but **uneven**: it clears the baseline on all three primary metrics (QWK 0.605 versus 0.537) by strengthening the well-populated upper-engagement region and the middle disengaged class, but the ∼2–3%-frequency *Highly-Disengage* class remains the principal open challenge (RQ1). This is the in-domain picture; the next section shows it becomes more severe still once the model leaves its training corpus — and, at the same time, that the architecture's advantage over the baseline *widens* there.

5.6 Cross-Dataset Generalization

The decisive question is whether the hybrid's in-domain advantage survives, and pays off, on data the model was never trained on. Chapter 4 established with the baselines that engagement models do not transfer across corpora, that accuracy masks the failure, that pooling source corpora is the most effective simple lever, and that in-domain rank barely predicts transfer. This section first confirms the collapse on the hybrid, then establishes the claim that completes the thesis's contribution: **the hybrid generalizes better than the baselines** — in every cell, and most of all on the self-collected private set.

5.6.1 Cross-Dataset Matrix

The hybrid cross-dataset heatmap (Figure 5.9) shows the same pattern as the baseline matrix of Section 4.5: a healthy diagonal and an off-diagonal collapse of all three primary metrics — the confirmation anticipated in Section 5.2. The hybrid does not rescue cross-corpus transfer.

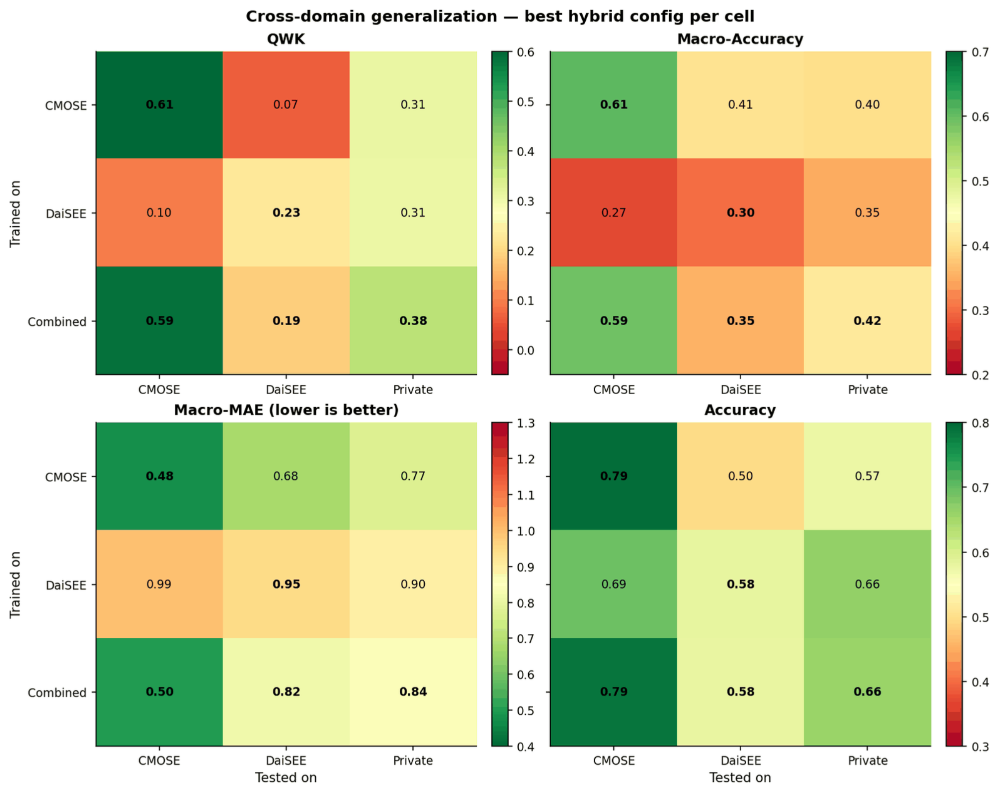


Figure 5.9: Cross-domain generalisation, best hybrid per cell, on the three primary metrics — QWK, macro-accuracy, and macro-MAE (lower is better) — plus raw accuracy, which again stays high in the off-diagonal cells where the chance-corrected metrics collapse.

The accuracy deception persists. On DAiSEE→CMOSE the best QWK any of the 486 hybrid runs achieves is 0.10, yet the accuracy panel of Figure 5.9 shows hybrids in that cell posting accuracies up to 0.69; on CMOSE→DAiSEE the best QWK is 0.07 against accuracies near 0.50. A second architecture family, developed independently of the baselines, thus lands in the same place: accuracy reports a partial success in exactly the cells where the chance-corrected metrics report chance-level agreement. The metric frame of Section 4.2.1 is therefore not an artefact of the monolithic models — and this is the last time accuracy appears in the thesis's analysis.

Within the collapse, the hybrid wins every cell. Comparing Figure 5.9 with Figure 4.10 cell by cell, the best hybrid beats the best baseline in **all nine train×test cells**: on the diagonals (0.605 vs 0.537 on CMOSE; 0.228 vs 0.166 on DAiSEE), in the collapsed cross-corpus cells (e.g. 0.096 vs 0.045 on DAiSEE→CMOSE), and across the whole private column (0.309 vs 0.190 from CMOSE; 0.306 vs 0.256 from DAiSEE; 0.379 vs 0.285 from Combined). The improvement is not a single tuned point but a uniform shift of the entire matrix — the strongest form the family-level claim can take.

5.6.2 The Private Set

The most consequential evaluation in this thesis is on the **private set** — the 366 clips we collected and hand-labelled ourselves (Section 3.2.2), processed through the identical OpenFace/I3D pipeline and used **for testing only**. Because no model is ever trained on it, the private set is a genuine "in-the-wild" probe of deployment: it has its own recording conditions and its own label prior (58% Engage, 31% Highly-Engage, only 2.7% Highly-Disengage). For this set there is no in-domain option at all; the *only* design lever is the choice of training source, which makes it the cleanest test of the two contributions of this thesis acting together — the semantic-group architecture and the corpus-pooling insight of Section 4.5. Figure 5.10 and Table 5.4 report the best base model and the best semantic-group hybrid on the private set, for each training source.

Table 5.4: T6. Private set (test-only): best base vs best hybrid by training source (macro-MAE lower is better).

Train source	Best base (model/loss)	Base QWK	Base macro-MAE	Best hybrid (arch)	Hybrid QWK	Hybrid macro-acc	Hybrid macro-MAE
CMOSE	openface_transformer/ce	0.190	0.750	Hybrid (OpenFace only) TCN_T_LSTM_TCN_T	0.309	0.385	0.914
DaiSEE	i3d_mlp/ce	0.256	1.031	Hybrid + I3D LSTM_LSTM_TCN_LSTM_LSTM	0.306	0.350	0.966
Combined	openface_transformer/ce	0.285	0.997	Hybrid + I3D T_T_T_LSTM_T	0.379	0.385	0.877

Two effects compound, and both favour this thesis's contributions.

1. **Training source matters, and the combined corpus wins.** For both model

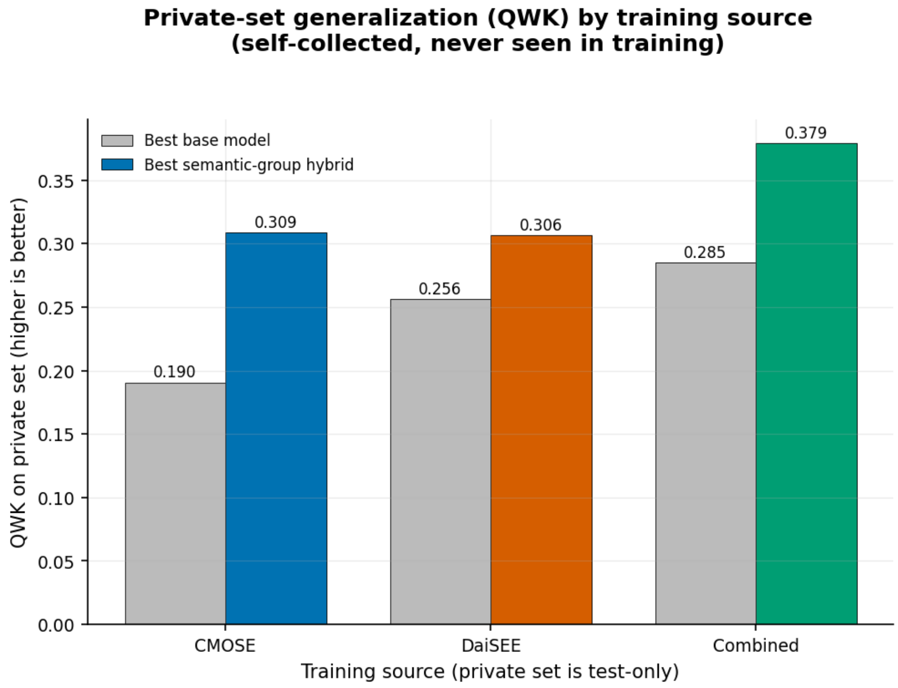


Figure 5.10: Private-set generalisation by training source: best base versus best hybrid, on QWK (higher is better) — the headline generalisation metric. Macro-accuracy and macro-MAE for the same models are in Table 5.4.

families, training on **Combined** generalises best to the private set; a single source is markedly worse (best base QWK: Combined 0.285 versus DAiSEE 0.256 versus CMOSE 0.190). Pooling heterogeneous source corpora is the most effective simple route to an unseen target, exactly as the cross-dataset matrix of Chapter 4 predicted.

2. **The hybrid generalizes better — and helps more here than in-domain.** The semantic-group hybrid beats the best base model on QWK for *every* training source, and the gap is **largest exactly where it matters**: under combined training the hybrid reaches **QWK 0.379 versus 0.285** for the best base, a $+0.094$ improvement — wider than the $+0.068$ in-domain CMOSE gain (Section 5.5). This is the crucial new claim of the section: the architecture's advantage is not an in-domain artefact that distribution shift erodes, it is a generalization property that distribution shift *amplifies*. The overall best private-set model is the combined-trained I3D-fused hybrid `T_T_T_LSTM_T`, and it is best on both ordinal primaries: **QWK 0.379** and the lowest **macro-MAE 0.877** of any source (Table 5.4), against the best base's 0.997. Macro-MAE corroborates the QWK ranking with one honest caveat: the CMOSE-only base model attains a deceptively low macro-MAE (0.75) despite the weakest QWK (0.19), because it predicts conservatively around the majority classes — precisely the kind of false comfort that reporting all three primaries together is designed to catch.

In other words, the architecture contribution and the combined-training insight are not independent curiosities: they **stack**, and they stack most strongly on the hardest, most realistic, self-collected target.

Figure 5.11 shows the row-normalised confusion of this headline model on the private set, the counterpart, under distribution shift, to the in-domain analysis of Section 5.5. The ordinal structure is captured (mass concentrates on the diagonal and its neighbours): *Engage* recall is 0.72 and *Highly-Engage* 0.53, with errors mostly to adjacent classes. As in-domain, the rare disengaged classes are the failure mode under shift — *Disengage* recall is only 0.29 and **Highly-Disengage collapses to 0.00** (all ten private HD clips are misread as DE/EG).

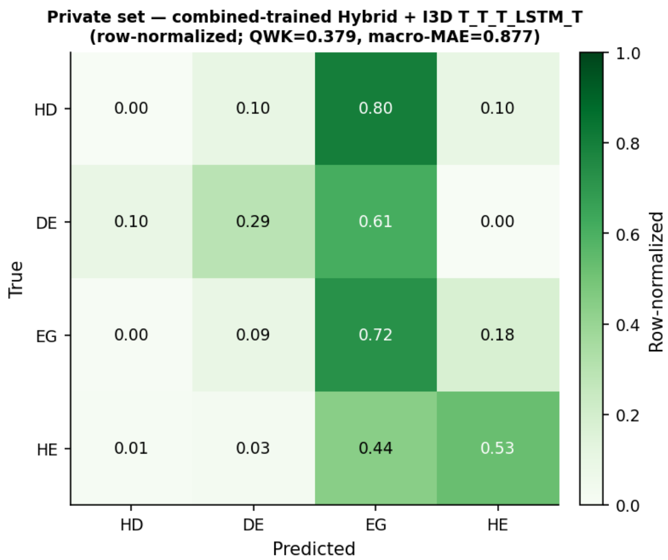


Figure 5.11: Private-set confusion, combined-trained I3D hybrid `T_T_T_LSTM_T` (QWK 0.379, macro-MAE 0.877), row-normalised.

5.6.3 In-Domain versus Transfer

Section 4.5.3 showed that a baseline's in-domain QWK barely predicts its transfer ($\rho = 0.28$); the ablation population sharpens that warning. Figure 5.12 overlays the 972 trained hybrid models on the baseline scatter: within the hybrid family the Spearman correlation between in-domain QWK and unseen-target QWK is **negative** ($\rho = -0.31$) — among closely related configurations, squeezing out more in-domain QWK mildly *hurts* transfer, because the extra fit is corpus fit. This puts the hybrid's generalisation advantage in its proper light: the architecture does not win by sitting further along an in-domain–transfer trend — no such usable trend exists — it wins by **lifting the whole trade-off**, sitting above the baseline cloud at every level of in-domain performance. It also yields a practical warning for model selection: picking the single best in-domain configuration is not how the private-set winner is found; family-level robustness and pooled training are.

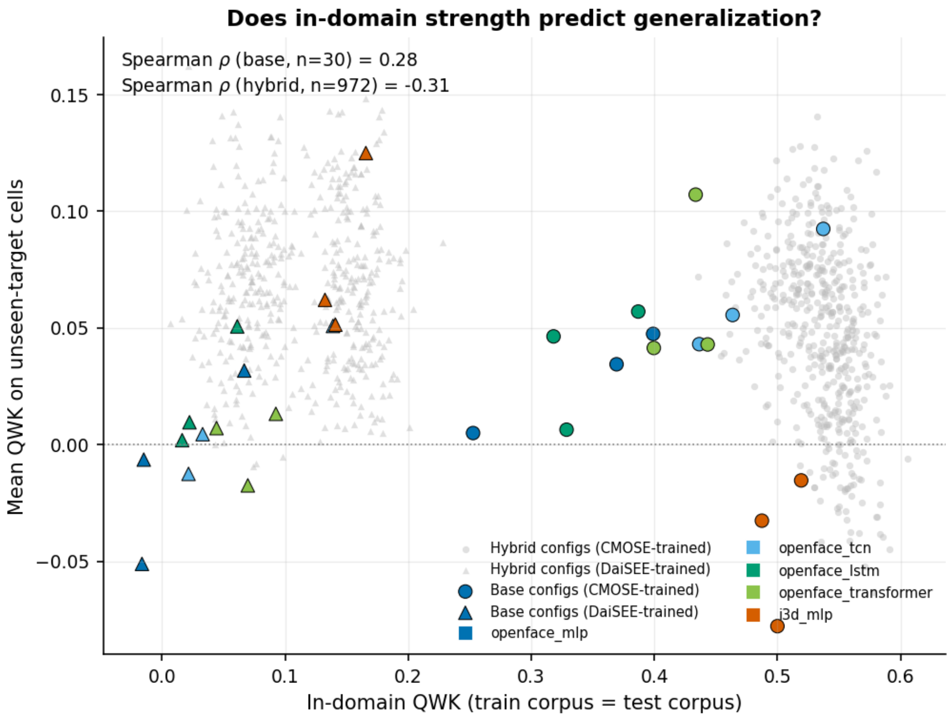


Figure 5.12: In-domain QWK versus mean unseen-target QWK with the hybrid ablation population (grey) overlaid on the baseline configurations (coloured). Within the hybrid family the correlation is negative ($\rho = -0.31$): more in-domain fit means more corpus fit. The hybrid cloud sits above the baselines throughout — the architecture lifts the trade-off rather than riding it.

5.7 Discussion

Why pooling is the effective lever. The cross-dataset matrix and the private-set result agree that training on the *combined* corpus generalises best, and the dataset statistics of Section 3.2 explain why. CMOSE and DAiSEE are imbalanced in opposite directions — CMOSE toward *Engage*, DAiSEE additionally toward *Highly-Engage* — so their union spans a wider range of label priors than either alone. An unseen target such as the private set, with its own distinct prior, is consequently closer in distribution to the pooled corpus than to either single source, and the model sees both upper-engagement regimes in training rather than overfitting one. Pooling is thus not a tuning trick but a direct consequence of prior coverage, which is why it helps with no change to the architecture or loss.

Why the hybrid helps more under shift. That the architecture's margin over the baseline *widens* off-domain ($+0.094$ versus $+0.068$) is consistent with the decomposition's design rationale. Encoding each behavioural channel separately, and forcing each to be discriminative through its auxiliary head, yields a representation less tied to any single corpus's idiosyncrasies than a monolithic encoder that can latch onto a corpus-specific shortcut. The negative in-domain–transfer correlation within the ablation ($\rho = -0.31$, Section 5.6.3) is the same mechanism seen from the other side: whatever a configuration gains *beyond* the family's level of in-domain fit is corpus fit, and it is the structured family — not the most tuned member — that carries the transfer advantage.

Why the two feature families behave differently under shift. The paired I3D ablation (Section 5.4) and the baseline architecture ranking (Section 4.5.1) tell one story at two scales: the OpenFace behaviour descriptors — gaze angles, head-pose rotations, action-unit intensities — are defined in subject-centred physical units that mean the same thing in any recording setup, whereas the I3D embedding encodes global appearance, which is exactly what changes between corpora. Fusion therefore adds signal where the appearance statistics of the test data are represented in training, and adds bias where they are not.

What the headline number does and does not mean. The best private-set result, QWK 0.379, should be read honestly. It is a clear improvement over chance, over every single-source model, and over the best baseline ($+0.094$), and it shows the architecture and pooling contributions stacking on unseen data. It is also, in absolute terms, *moderate*: a QWK below 0.4, together with a macro-MAE of 0.877 — almost a full ordinal step of average per-class error — means the model captures the broad ordinal trend — engaged versus disengaged — but is unreliable at fine distinctions and, as the confusion matrix shows, blind to the rarest disengaged class. For deployment this implies a system usable as an aggregate, classroom-level engagement signal, but one that should not be trusted to flag an individual highly-disengaged learner without human confirmation.

The persistent failure mode. Across in-domain and shifted evaluation the same class fails: *Highly-Disengage*, the rarest (under 3%) and the most decision-relevant. In-domain it is traded away (Section 5.5); under shift it collapses to zero recall. This is the clearest sign that the open problem is not architectural but one of data and objective: no encoder assignment can recover a class the training distribution barely contains, which is precisely why the imbalance-specific remedies and richer data proposed in Chapter 6 are the natural next step.

5.8 Chapter Summary

This chapter evaluated the semantic-group hybrid under the frame fixed in Chapter 4 — after first re-validating that frame on the 486-configuration ablation population, where the metric block structure, the blindness of accuracy, and the cross-domain collapse all reproduced, establishing them as task properties (Section 5.2). Only head pose has a clear encoder preference — a TCN — and the preference survives domain shift, so the design space is mostly flat and the benefit comes from decomposing and fusing rather than from the encoder search (Section 5.3, RQ2). The paired I3D ablation localised the fusion effect by regime: decisively positive where the test corpus is seen in training (+0.060 QWK, 94% of pairs; 82% of fused configurations clear the baseline bar in-domain), neutral-to-harmful beyond it (RQ3) — the family-level confirmation of Chapter 4's feature-complementarity analysis on the seen regime, and of its I3D rank-drop under shift (Section 5.4). In-domain the best hybrid beats the best baseline on all three primary metrics (+0.068 QWK, 0.605 versus 0.537), with the gain concentrated in the upper-engagement region and Highly-Disengage the remaining open problem (Section 5.5, RQ1). The generalization study (Section 5.6) established the new result that completes the thesis's argument: the hybrid beats the best baseline in *every* cell of the 3×3 matrix, compounding with combined-corpus training to the best private-set model (QWK 0.379, lowest macro-MAE 0.877, $+0.094$ over the baseline — a wider margin than in-domain), while the in-domain–transfer correlation within the family is negative ($\rho = -0.31$): the architecture lifts the whole in-domain–transfer trade-off rather than riding it (RQ4). In both regimes the rare Highly-Disengage class remains the principal failure mode, collapsing entirely under distribution shift. Chapter 6 draws the findings together, revisits the research questions, and outlines future work.

6.1 Summary

This thesis studied four-class ordinal student-engagement recognition from per-frame OpenFace facial-behaviour descriptors and clip-level I3D motion features. It set out to improve recognition over standard monolithic models and, equally, to characterise the problem honestly — in particular its severe, heterogeneous class imbalance and its behaviour under distribution shift.

Methodologically, the thesis built a complete and reproducible pipeline: feature extraction and leakage-free preprocessing; a **semantic-group hybrid temporal architecture** that decomposes the 709-dimensional OpenFace descriptor into five behavioural groups (gaze, eye landmarks, face landmarks, head pose, action units), encodes each group with its own temporal encoder, and optionally fuses a global I3D stream while supervising every stream with an auxiliary head; five monolithic baselines; three loss functions; and a 3×3 cross-dataset evaluation protocol. The central idea was validated by training and evaluating all $3^5 = 243$ per-group encoder assignments, with and without I3D, and by a **self-collected, test-only private set** of 366 hand-labelled clips that serves as the decisive deployment probe.

The evidence supports five conclusions.

1. **The I3D-fused hybrid is the best in-domain model and, as a family, beats the strongest monolithic baseline.** The I3D-fused family is centred above the baseline QWK of 0.537 (median 0.553, with 82% of its 243 configurations clearing the baseline), and the best configuration reaches **QWK 0.605** with a mixed TCN/Transformer/LSTM assignment (`TCN_T_TCN_LSTM_T`). The gain over the best baseline is genuine but **modest** ($+0.068$ QWK); the value lies in the framework and the systematic evidence, not in a leap in state of the art.
2. **Most groups are encoder-agnostic; only head pose shows a clear preference, favouring a TCN** (a 0.022 QWK spread across the three encoders, against ≤ 0.006 for the other four groups). This is an interpretable design rule reflecting the temporally local nature of head dynamics.
3. **Engagement classifiers do not generalise across datasets.** Off-diagonal QWK collapses to near zero (CMOSE→DAiSEE 0.02) even when accuracy looks acceptable, so ordinal agreement metrics — not accuracy — must be reported on imbalanced engagement data.

4. **On the self-collected, hand-labelled private set — the most realistic test — the two contributions compound.** The hybrid's advantage is not confined to one comparison: it beats the best baseline in *all nine* train$\times$test cells of the cross-dataset matrix. Because the private set is test-only, the only lever is the training source; training on the **combined** corpus with the **semantic-group hybrid** gives the best result (QWK 0.379), beating the best baseline by $+0.094$ (against $+0.068$ in-domain) and topping every single-source hybrid. This is the practical payoff of the thesis.
5. **The rare Highly-Disengage class remains the principal open problem.** It is around 2–3% of the data, is traded away rather than solved by the hybrid in-domain, and collapses to zero recall under distribution shift.

A note on scope: the architecture ablation is anchored on CMOSE because DAiSEE's in-domain ordinal signal is too weak (QWK 0.166) for architecture differences to be distinguishable from label noise; CMOSE is therefore used for development and the private set for the decisive validation.

6.2 Synthesis

Read together, the two results chapters tell a single story whose moral is not the one a benchmark paper usually reports. The thesis has a *constructive* side — the semantic-group hybrid and the corpus-pooling recipe — and a *critical* side — an honest account of imbalance, non-transfer, and the metrics that expose them — and the most important finding lies in how the two interact.

The architecture earns its value off-domain, not on it. In-domain the semantic-group decomposition is a modest, interpretable improvement: the I3D-fused family sits above the baseline but the best configuration clears it by only $+0.068$ QWK, and most of the 243-cell design space is encoder-agnostic. Taken alone, that is a small result. What makes it worth the construction is that the same decomposition pays off *more* exactly where it is hardest to: on the unseen private set the margin over the baseline widens to $+0.094$, and it compounds with pooling rather than competing with it. The contribution is therefore best understood not as a new in-domain state of the art but as a representation that degrades more gracefully under distribution shift — which, for a problem whose only real test is deployment, is the property that matters.

The fusion was measured before it was built. The decision to fuse the OpenFace and I3D families did not rest on intuition: the clip-level agreement analysis of Chapter 4 showed that the best OpenFace model and the I3D baseline disagree more with each other than OpenFace models do among themselves, and that each family solves clips the other cannot — evidence of complementary errors that the paired I3D ablation of Chapter 5 then confirmed in-domain. The same analysis also quantified untapped headroom: an oracle over the thirty baseline configurations would reach 97.6% of clips against 77.3% for the best single model, which both motivates fusion and marks ensemble selection as a direction the thesis deliberately left aside.

The flat design space is itself the message. That four of five behavioural groups are indifferent to their encoder, while only head pose prefers a TCN, bounds the contribution honestly: the gain comes from *decomposing and fusing* the descriptor, not from a clever per-group encoder search. This is a more transferable design rule than a single tuned configuration, and it is the reason the family-level effect, rather than any one cell, is reported as the result.

The negative results are contributions, not caveats. The near-total collapse of cross-corpus transfer, the way raw accuracy disguises it, and the persistent failure on the rarest disengaged class are not incidental limitations of these particular models; they are properties of the problem that the field's in-domain, accuracy-led habit leaves unmeasured. Establishing them quantitatively — and showing that QWK, macro-accuracy, and macro-MAE make them visible — reframes engagement recognition from "beat the benchmark" to "report ordinal agreement and instrument rare-class failure," which is the discipline the rest of this chapter recommends.

6.3 Revisiting the Research Questions

The four research questions posed in Section 1.3 can now be answered directly.

RQ1 (decomposition). *Yes, decomposition helps.* Encoding the five behavioural groups separately and fusing them outperforms encoding all 709 features with a single monolithic encoder: as a family the hybrid is centred above the best baseline, and the effect is robust rather than a single lucky configuration, with most of the I3D-fused configurations clearing the baseline.

RQ2 (encoder assignment). *Only head pose matters.* Four of the five groups are essentially indifferent to whether their encoder is a TCN, a Transformer, or an LSTM (spread ≤ 0.005 QWK); head pose alone shows a clear and interpretable preference for a TCN. The practical implication is that the design space is mostly flat and that a TCN is a safe default for every group, with head pose the one place the choice is worth making deliberately.

RQ3 (motion fusion). *Yes in-domain, with an honest caveat under shift.* When the model is evaluated on a distribution it has trained on, adding the clip-level I3D embedding is decisively beneficial: the paired comparison over all 243 OpenFace assignments shows a median gain of $+0.060$ QWK with 94% of pairs improving, and the fused family produces both the best in-domain model and the best private-set model. Under cross-corpus shift, however, the same stream is neutral to mildly harmful on average (median -0.021 on cross-public cells, $+0.005$ on the private set) — the I3D stream encodes useful but corpus-specific appearance cues. The fusion is therefore recommended, but its gain should be claimed only for the regime in which it holds.

RQ4 (generalisation and losses). *Generalisation is poor and metric choice is decisive.* Models do not transfer off-the-shelf across corpora; off-diagonal ordinal agreement collapses while accuracy stays deceptively high. In-domain merit is, moreover, a poor predictor of transfer: the correlation between in-domain and unseen-target QWK is weak for the baselines ($\rho = 0.28$) and mildly negative across the hybrid grid ($\rho = -0.31$), so a model cannot be selected for deployment from its in-domain score alone. Among losses there is no universal winner — cross-entropy maximises ordinal agreement (QWK) while weighted and ordinal losses recover minority recall and balanced ordinal error — so the loss is a Pareto choice; notably, the cross-entropy-first ranking is the one in-domain preference that is preserved intact on every unseen target, which retrospectively validates fixing it before development. QWK together with macro-accuracy and macro-MAE must be the primary yardstick, and pooling source corpora is the single most effective lever for an unknown target.

6.4 Limitations

The findings should be read against several limitations, each of which also motivates the future work below.

- **Single-annotator private labels.** The private set was labelled by one annotator (the author) under the public-corpus rubric. Although the model suggestions shown during labelling were non-binding, the absence of a second annotator means no inter-rater reliability can be reported for this set, and the 366-clip size limits the statistical power of the private-set conclusions — especially for the ten Highly-Disengage clips, where a single misjudged clip moves the recall substantially.
- **Clip-level I3D features.** CMOSE and DAiSEE ship a single pooled 1024-dimensional I3D vector per clip, so the I3D "stream" is an MLP over that global motion/appearance summary rather than a temporal encoder; a per-window I3D sequence might contribute more and would let an I3D temporal

encoder model genuine dynamics.

- **Incremental in-domain gain and a flat design space.** Most of the 243-configuration grid is encoder-agnostic, and the best in-domain improvement over the baseline is modest ($+0.068$ QWK). The decomposition is a sound, interpretable framework rather than a large performance leap.
- **Weak DAiSEE signal.** The very low in-domain QWK on DAiSEE prevented using it for architecture development and means the cross-dataset conclusions rest mainly on CMOSE and the private set.
- **Fixed training recipe.** A single seed, learning rate, and encoder size were used throughout; the absence of multi-seed repetition means the reported margins are point estimates rather than confidence intervals, and small differences between neighbouring configurations should not be over-interpreted.

6.5 Practical Recommendations

For practitioners building engagement-monitoring systems, the study yields three concrete recommendations that are independent of the specific architecture. First, **report and optimise ordinal agreement**: use QWK, macro-accuracy, and macro-MAE as the headline metrics and treat raw accuracy as secondary, because on these imbalanced ordinal labels accuracy can look healthy while the model has learned nothing useful about disengagement. Second, **pool heterogeneous source corpora** when the deployment distribution is unknown; pooling was the most effective simple lever for the unseen private set and requires no change to the model. Third, **expect and instrument rare-class failure**: the most decision-relevant class, Highly-Disengage, is also the rarest and the least reliably detected, so a deployed system should treat low-confidence disengagement predictions cautiously and, where possible, defer to a human.

6.6 Suggestion for Future Works

- **Targeting the rarest class.** Combine the architecture with imbalance-specific remedies — ordinal losses tuned for Highly-Disengage, minority oversampling such as SMOTE [15], or cost-sensitive decision thresholds — to recover HD recall without trading away Highly-Engage. Because HD is so rare, even modest gains here would change the practical value of the system more than further QWK gains on the frequent classes.
- **Domain adaptation.** The cross-dataset gap invites explicit unsupervised domain adaptation or domain-invariant representation learning, rather than the naive corpus pooling that already helps; a small amount of labelled target data could

be used for calibration before deployment.

- **Learned, not enumerated, group encoders.** Replace the discrete 243-configuration search with a differentiable architecture search or a gating mechanism that selects per-group temporal operators end-to-end, removing the need to enumerate the design space and allowing the model to discover group–encoder affinities automatically.
- **Stronger and richer motion features.** Re-extract per-window I3D (or a modern video backbone) so the motion stream carries temporal dynamics, and fuse audio/speech features, which CMOSE also provides, as additional semantic streams within the same decompose-then-fuse framework.
- **Frequency-domain analysis of engagement (exploratory).** A natural complementary direction — not implemented in this thesis and left strictly as future work — is to analyse the *temporal frequency* of facial signals, under the hypothesis that engaged behaviour is low-frequency and stable while disengaged behaviour is higher-frequency and fidgety, for example via short-time-Fourier-transform features or convolutions over the frequency axis.
- **Stronger evaluation.** Multi-annotator labelling of a larger private set, with reported inter-rater agreement, and multi-seed training with confidence intervals would tighten every margin reported here and turn point estimates into statistically grounded claims.

6.7 Closing Remarks

Automatic engagement recognition is attractive precisely because online learning makes the webcam the only window onto the learner, but the same setting makes the problem hard: the labels are ordinal, severely imbalanced, subjective, and corpus-specific. This thesis showed that respecting the structure of the facial descriptor — dividing it into behavioural groups and encoding each on its own terms — yields a sound and interpretable model that, fused with a motion stream and trained on pooled corpora, generalises best to the most realistic, unseen data. Just as importantly, it showed that the field's habitual reliance on accuracy hides a near-total failure to transfer across datasets, and that ordinal agreement metrics expose it. Recognising the rarest, most decision-relevant disengaged behaviour under distribution shift remains open, and is, in the author's view, the most worthwhile next step.

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